Unit 47 Industrial processes

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

47.1 Building a chemical plant in a city

The following is an excerpt from a newspaper article. Three industrial companies intend to build plants in City X.

City X Newspaper
Which company? All of them may spell jobs for us

Several weeks ago, the city council announced that three industrial companies are interested in building chemical manufacturing plants in our city —

AA Nitrogenous Fertilizer Company
BB Chlorine Gas Company
CC Methanol Company

Our economy went down since the bankruptcy of the XX Bank last year. Many residents have lost their jobs as so many companies have closed.

Can anything be done to boost our local economy? Yes. The answer lies in allowing one of these companies to build a chemical plant in our town. New jobs would be created..... However, we must question the potential safety and environmental risks that each of these chemical plants would pose.....

(Source: JKP 03/2008)

Suppose you were a citizen in the city. What is your decision? Let us first look into these three industries.



Fig. 47.1 Nitrogen is needed for proteins for healthy leaves

N1 Refer to page T46.

Animation
Nitrogen cycle

47.2 What do plants need to grow properly?

Plants need water from the soil. They also need other elements. The three elements that they need most are nitrogen, phosphorus and potassium. All plants and animals contain nitrogen in amino acids, polypeptides, proteins and nucleic acids. Plants cannot obtain nitrogen directly from the air. Instead, they take up nitrogen-containing compounds from the soil. N1

The nitrogen cycle describes the ways that nitrogen and its compounds circulate within the environment and living organisms. Processes that convert atmospheric nitrogen into nitrogen-containing compounds are a very important part of the cycle. They are known as **nitrogen fixation** processes.

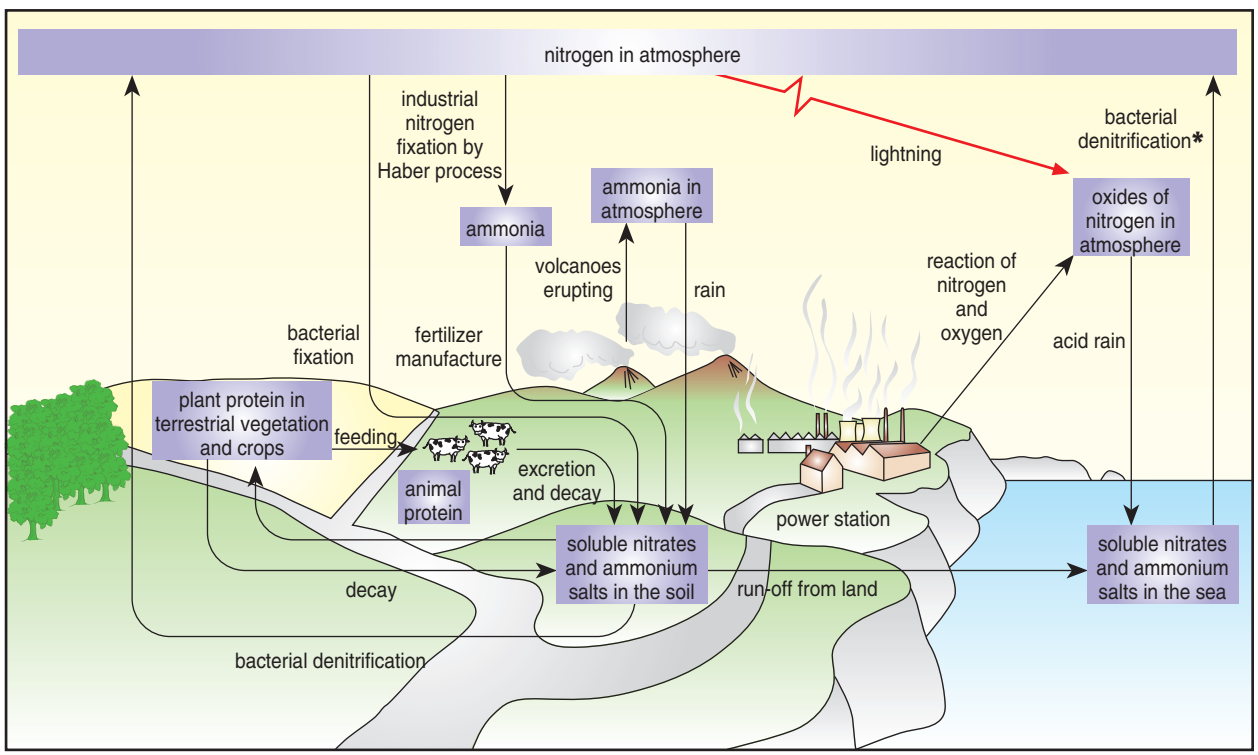


Fig. 47.2 The nitrogen cycle

* Denitrifying bacteria live in heavy, wet soils. They can break down nitrogen-containing compounds, releasing nitrogen back into the air.

nitrogen fixation 固氮作用

Nitrogen fixation processes

There are three main ways of fixing nitrogen from the air:

- a chemical reaction in the air during lightning flashes;
- the action of bacteria;
- the manufacture of ammonia by the Haber process.

CE 2000 Paper 1 Q9(c)(ii)

Lightning can provide sufficient energy for the direct combination of nitrogen and oxygen in the air to form nitrogen monoxide.

$$N_2(g) + O_2(g) \longrightarrow 2NO(g)$$

Some plants, such as peas and clovers, are able to convert atmospheric nitrogen into ammonium ion by the action of bacteria found in root *nodules*. The ammonium ion may be oxidized by *nitrifying bacteria* to give nitrate ion.



Fig. 47.3 Lightning makes nitrogen and oxygen in the air combine to form nitrogen monoxide



Fig. 47.4 Bacteria found in these root nodules are able to convert atmospheric nitrogen into ammonium ion

Growing crops take in the nitrogen they need from the soil in the form of nitrate. There must be a constant supply of nitrogen-containing compounds to the soil for plants to grow healthily. Much of the nitrogen cycle has taken place naturally for many millions of years.

nodule 小節　　nitrifying bacteria 硝化細菌

* Intensive farming is a method of farming that uses large amounts of capital or labour to obtain the maximum yield from a limited area of land.

However, *intensive farming** removes more nitrogen-containing compounds than are naturally put back into the soil, so it has become necessary to supplement the supply of nitrogen by adding nitrogenous fertilizers.



Fig. 47.5 Intensive farming on a big scale

* Land areas are conveniently measured in hectares (ha).
1 ha = 1 x 10^4 m^2
A football pitch is about 0.5 hectare.

Many farmers rely on commercial nitrogenous fertilizers. The active ingredients typically are inorganic compounds that contain nitrogen, phosphorus and potassium. These fertilizers account for about one-fourth of the world's crop yield.

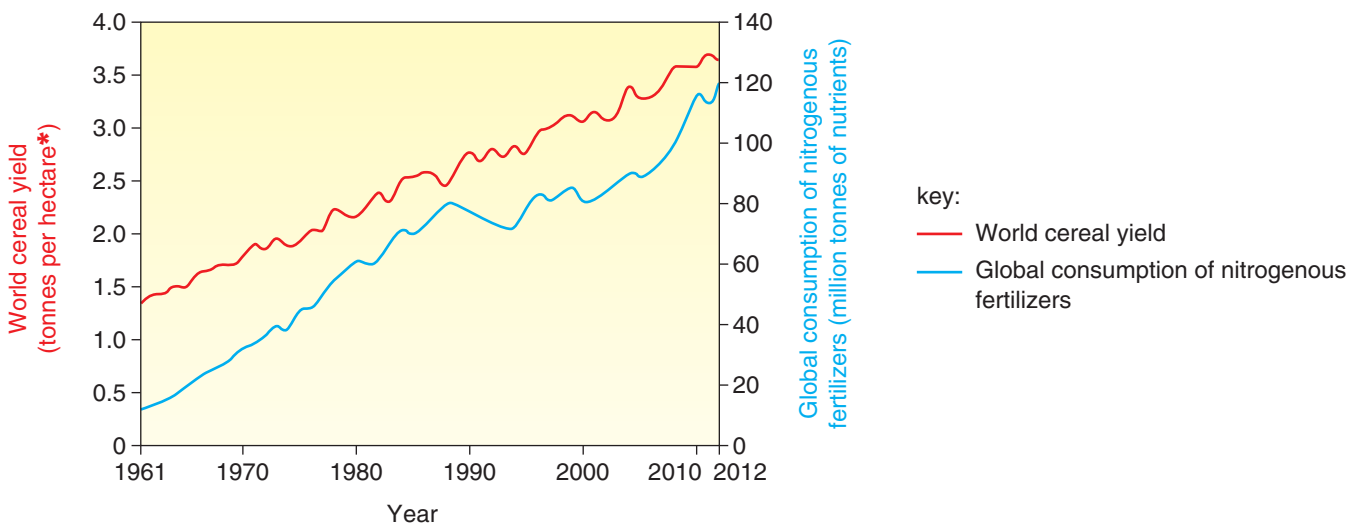


Fig. 47.6 World cereal yield shown against global consumption of nitrogenous fertilizers (1961 – 2012) (Source: FAOSTAT)

intensive farming 集約農業

47.3 Nitrogenous fertilizers

Nitrogenous fertilizers contain chemicals such as

- ammonium nitrate;
- ammonium sulphate;
- ammonium phosphate;
- triple superphosphate (a form of calcium phosphate);
- urea ($CO(NH_2)_2$); and
- potassium chloride.

There are two main types of nitrogenous fertilizers:

- Straight N fertilizers for farmers who want only nitrogen (N). These fertilizers usually contain ammonium salts.
- NPK compound fertilizers for farmers who want nitrogen (N), phosphorus (P), and potassium (K). These fertilizers contain the elements in varying amounts to suit different soils and crops.



Fig. 47.7 NPK fertilizer contains nitrogen (N), phosphorus (P) and potassium (K)

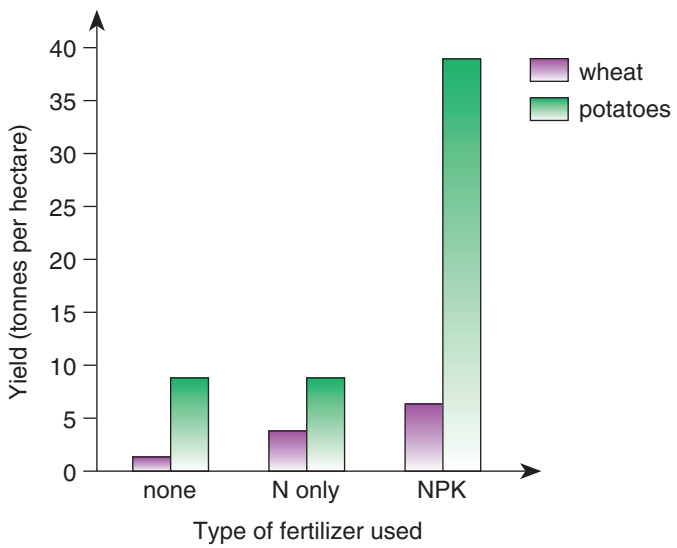


Fig. 47.8 Each crop has its own nutrient needs

Ammonia is a feedstock for making nitric acid. Almost 80% of the world production of nitric acid is used to manufacture ammonium nitrate, and most of this is used as fertilizer.

CE 2002 Paper 2 Q42
DSE 2014 Paper 2 Q1(c)(i)
AL 2006 Paper 1 Q9

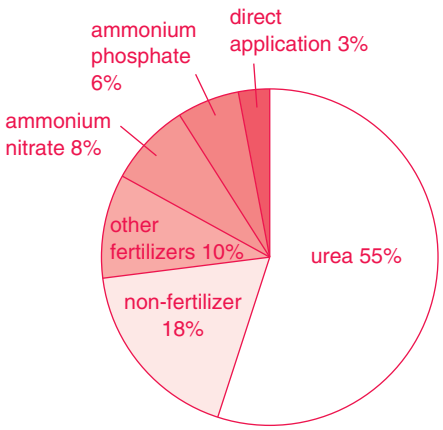
47.4 Ammonia — the key to nitrogenous fertilizers

Virtually all ammonia produced today is made by the Haber process. The key step of the process is the direct reaction of nitrogen and hydrogen:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92 \text{ kJ mol}^{-1}$$

AL 2006 Paper 1 Q9

N2 End uses of ammonia produced (2012)



(Source: Merchant Research & Consulting Ltd., 2013)

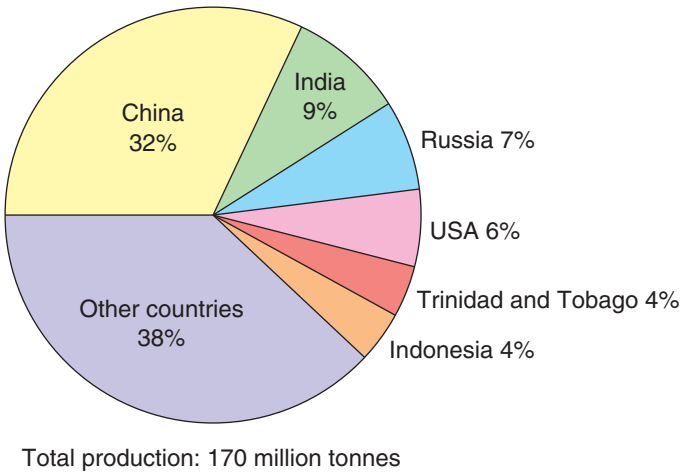


Fig. 47.9 Global ammonia production 2012 N2
(Source: US Geological Survey, Mineral Commodity Summary)

Feedstocks

CE 2000 Paper 2 Q10
DSE 2014 Paper 2 Q1(c)(ii)

Nitrogen can be obtained by the fractional distillation of liquid air, while hydrogen can be obtained from natural gas. N3

N3 Air cannot be used instead of nitrogen in the catalytic converter because air contains oxygen. A mixture of oxygen and hydrogen is explosive.

To produce the hydrogen gas needed, natural gas is first treated to remove sulphur-containing compounds; then the methane present is allowed to react with steam:

DSE 2012 Paper 2 Q1(a)(ii)

$$CH_4(g) + H_2O(g) \longrightarrow 3H_2(g) + CO(g)$$

The ratio of methane to steam must be controlled carefully to prevent the formation of other carbon compounds.

The carbon monoxide from the above reaction is converted to carbon dioxide. Additional hydrogen gas is also produced.

$$CO(g) + H_2O(g) \longrightarrow H_2(g) + CO_2(g)$$

ASL 1999 Q14
AL 2006 Paper 1 Q9

Reaction conditions N4

N4 Examination questions often ask about the reaction conditions of the Haber process.

Fig. 47.10 shows a flow diagram that outlines the Haber process.

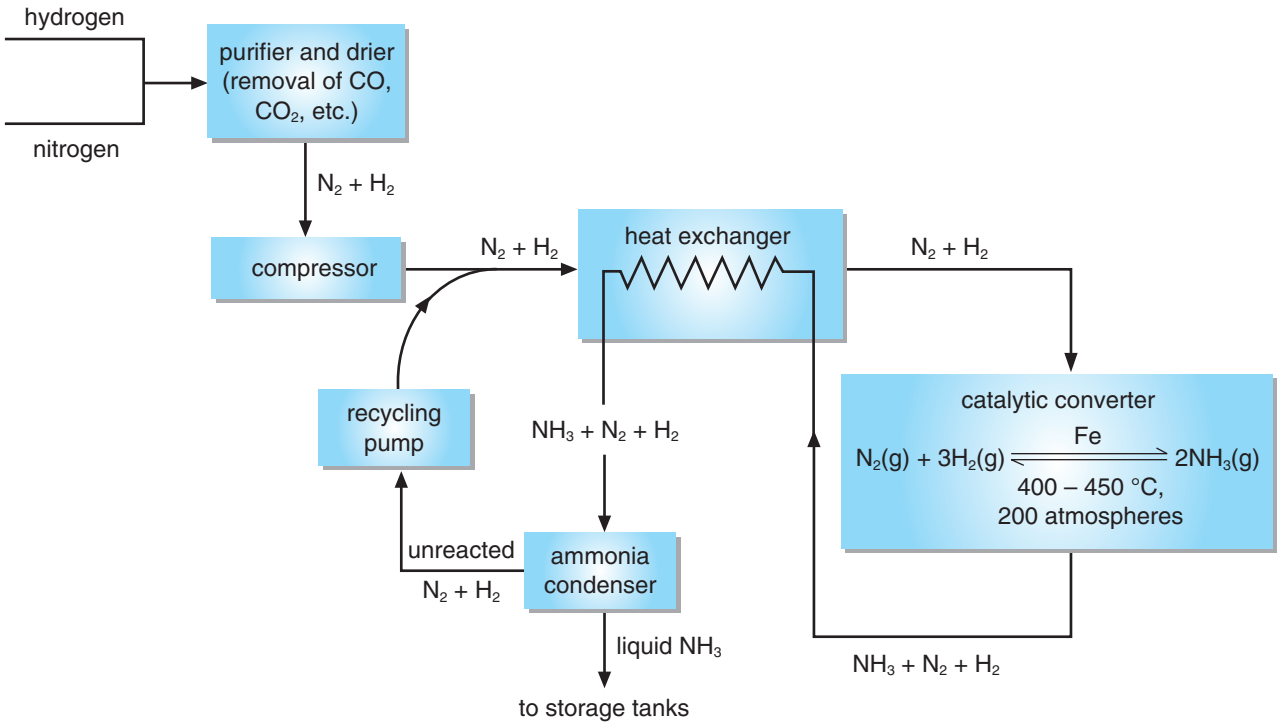


Fig. 47.10 A flow diagram that outlines the Haber process

ASL 1999 Q14
AL 2006 Paper 1 Q9

Thorough purification of both the nitrogen and hydrogen is necessary. This removes carbon dioxide and unreacted methane from the hydrogen. Impurities would poison the catalyst in the converter.

Reference Website
Refer to the following website for an animation showing the manufacture of ammonia in the Haber process:
http://www.absorblearning.com/media/item.action?quick=128
(accessed May 2015)



Fig. 47.11 An ammonia plant

CE 1999 Paper 1 Q8(b)(i)(1)
CE 2003 Paper 1 Q6(b)(ii)
DSE 2012 Paper 2 Q1(a)(i)(1)

N5 The product mixture is removed from the catalytic converter before reaching the yield of about 20%. This is because
- it takes a long time to attain equilibrium that the $NH_3(g)$ is of the highest yield;
- it can increase the total amount of $NH_3(g)$ produced per unit time.

CE 2000 Paper 2 Q10
CE 2002 Paper 2 Q25
ASL 1999 Q14
ASL 2010 Paper 1 Q11
AL 2006 Paper 1 Q9

Nitrogen and hydrogen are mixed in the mole ratio of 1 : 3. The mixture is passed over an iron catalyst in the catalytic converter at a pressure of 200 atmospheres. The reaction takes place at about 400 – 450 °C. N5

ASL 1999 Q14
ASL 2010 Paper 1 Q11
AL 2006 Paper 1 Q9

CE 1999 Paper 1 Q8(b)(i)(2)

Hot product gases are used to warm up the nitrogen and hydrogen in a heat exchanger before they enter the converter.

CE 2003 Paper 1 Q6(b)(iv)
DSE 2014 Paper 2 Q1(c)(v)

The ammonia is then cooled and condensed. The unreacted nitrogen and hydrogen are recycled.

Animation
Haber process

47.5 Obtaining an economic yield in the Haber process

The choice of temperature, pressure and catalyst are the conditions most likely to affect the yield of the chemical process:

DSE 2015 Paper 2 Q1(a)(ii)

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92 \text{ kJ mol}^{-1}$$

Here we shall look at each of them in turn.

Temperature

As Le Chatelier's principle suggests, because the forward reaction is exothermic, a low temperature will cause the position of equilibrium to shift to the right. This favours the formation of ammonia.

In choosing the right temperature, chemists must consider

- the percentage of ammonia in the equilibrium mixture (i.e. the yield);
- the rate of reaction.

The rate of reaction is related to temperature. If the temperature is too low, the reaction proceeds too slowly.

DSE 2014 Paper 2 Q1(c)(iv)

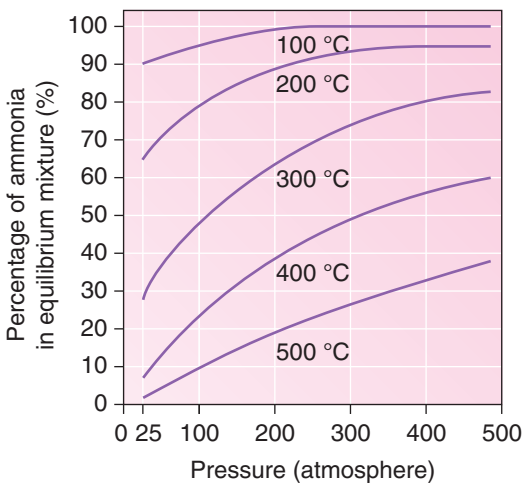


Fig. 47.12 Percentage of ammonia in the equilibrium mixture as temperature and pressure change

Pressure

As Le Chaterlier's principle suggests, a high pressure will cause the position of equilibrium to shift to the side with a smaller number of moles of gases. This favours the formation of ammonia.

A high pressure also increases the rate of a gas-phase reaction. However, using a high pressure drastically increases the cost of building and running the plant. Thicker pipes that can withstand high pressure are required to carry the gaseous mixtures and much more energy is used to maintain the pressure.

Reaction conditions most manufacturers use

Consideration of the factors leads to a 'compromise' set of conditions that most manufacturers use:

- a temperature between 400 and 450 °C — this is a compromise temperature at which the rate of reaction is high enough to give a reasonable yield in a short time.

DSE 2012 Paper 2 Q1(a)(iii)

- a pressure of about 200 atmospheres — low pressures will cause the position of equilibrium to shift to the left and thus decrease the yield of ammonia, while high pressures will increase the cost of building and running the plants; 200 atmospheres is a compromise.
- an iron catalyst — the catalyst speeds up both the forward and backward reactions so that the state of equilibrium is attained faster.

ASL 2011 Paper 2 Q6(c)

An industrial catalyst catalyzes a process by providing an alternative pathway with a lower activation energy. Thus, it can reduce the time for production and save energy of a process.

The cost of the process is also reduced by

- recycling the unreacted reactants;
- using the hot product gases to warm up the reactants.

47.6 From ammonia to nitrogenous fertilizers

The manufacture of nitric acid from ammonia involves three stages*:

* We have discussed this manufacturing process in Topic 12 Patterns in the Chemical World.



Fig. 47.13 This nitric acid plant is built close to an ammonia plant. Can you explain why?

Stage A Catalytic oxidation of ammonia to nitrogen monoxide (NO)

$$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g) \qquad \Delta H = -909 \text{ kJ mol}^{-1}$$

The conditions used are 900 °C, about 10 atmospheres, and a platinum-rhodium catalyst.

Stage B Oxidation of nitrogen monoxide (NO) to nitrogen dioxide (NO_2); dinitrogen tetroxide (N_2O_4) is formed upon cooling the nitrogen dioxide

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g) \qquad \Delta H = -115 \text{ kJ mol}^{-1}$$

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -58 \text{ kJ mol}^{-1}$$

Low temperatures (about 40 °C) and 7 – 12 atmospheres are used.

Stage C Reaction of dinitrogen tetroxide with water to form nitric acid

$$3N_2O_4(g) + 2H_2O(l) \rightleftharpoons 4HNO_3(aq) + 2NO(g) \qquad \Delta H = -103 \text{ kJ mol}^{-1}$$

(cold water)

CE 1999 Paper 1 Q8(b)(iii)

Ammonium nitrate is made by neutralizing nitric acid with ammonia solution:

$$HNO_3(aq) + NH_3(aq) \longrightarrow NH_4NO_3(aq)$$

CE 2001 Paper 1 Q8(b)
AL 2000 Paper 2 Q2(e)(i)
AL 2006 Paper 1 Q9



Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

1 The table below lists some information about four nitrogen-containing compounds which can be used as fertilizers.

Compound	Percentage by mass of nitrogen in compound	Solubility in water at 25 °C (mol dm^{-3})
NH_3	82.4%	31.1
NH_4NO_3	35.0%	26.8
$(NH_4)_2SO_4$	21.2%	5.8
$(NH_4)_2HPO_4$	21.2%	4.4

2 a) When the temperature is increased, the system will respond by reducing the temperature.
As the backward reaction is endothermic, the system will undergo a net backward reaction.
Thus, the yield of $N_2O_4(g)$ will decrease.

a) Suggest why most farmers do NOT prefer using ammonia (gas or solution) directly as a fertilizer. Ammonia is too alkaline and this makes the soil unsuitable for plants to grow.
Ammonia is a toxic gas.
Ammonia is volatile and difficult to store.

b) Suggest why ammonium nitrate is a widely used dry and granular form of nitrogenous fertilizer. (You need NOT consider the cost here.) Ammonium nitrate is very soluble in water and thus easily taken in by plants.
Ammonium nitrate has a high percentage by mass of nitrogen. The higher the percentage by mass of nitrogen in a fertilizer the better because less material needs to be stored and transported.

2 Consider the formation of $N_2O_4(g)$ from $NO_2(g)$ in the manufacture of nitric acid from ammonia.

$$2NO_2(g) \rightleftharpoons N_2O_4(g) \qquad \Delta H = -58 \text{ kJ mol}^{-1}$$

a) Describe the difference in the yield of $N_2O_4(g)$ you would expect at 40 °C and at 75 °C, with all other conditions being the same.

b) Le Chatelier's principle suggests using high pressure to convert $NO_2(g)$ to $N_2O_4(g)$. What is the advantage of using a lower pressure?
The cost of building and running the plant is lower.

Extension

47.7 Working on large scale production of nitrogenous fertilizers



Fig. 47.14 A student is deciding to use which reagents to prepare a sample of nitrogenous fertilizer

Imagine that you need to prepare a sample of nitrogenous fertilizer in the laboratory. You probably think of preparing either ammonium nitrate or ammonium sulphate. In either case, you may first neutralize a sample of ammonia solution with nitric acid or sulphuric acid, and then obtain the ammonium salt by crystallization.

$$NH_3(aq) + HNO_3(aq) \longrightarrow NH_4NO_3(aq)$$

$$2NH_3(aq) + H_2SO_4(aq) \longrightarrow (NH_4)_2SO_4(aq)$$

The procedure is quite simple for a laboratory preparation. How about producing enough fertilizers to meet the huge annual demand of a country?

Now imagine that you are a chemist and need to report on a possible process to produce say 100 000 tonnes of the fertilizers from ammonia per year. Here are some of the issues you probably need to consider.

Extension

Storage of raw materials

In the laboratory preparation, we can simply collect the ammonia solution and the required acid from reagent bottles. Such details cannot be left unmentioned when tonnes of materials are needed every day.

It is important to make planned decisions about the amounts of the raw materials to store. The following factors must be considered:

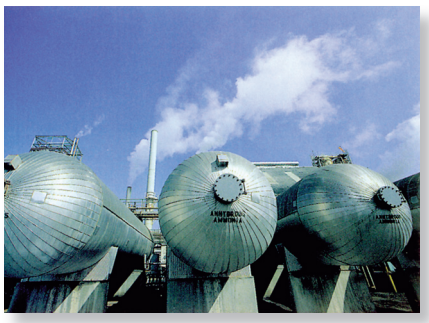


Fig. 47.15 Ammonia storage tanks

- the cost of the storage tanks, especially if the materials are corrosive, highly flammable, or toxic;
- the value of the land required for storage;
- the value of the materials stored and the working capital tied up with them;
- some materials may require special storage conditions, such as refrigeration, or storage out of contact with the air;
- additional safety concerns where materials are stored in bulk;
- economies of scale when buying in bulk — it is probably cheaper to buy in bulk rather than in small amounts;
- buy raw materials just when required rather than keeping stock (less storage space is required);
- the cost to the company should stocks run out and production be brought to a halt (if this occurs, the company may need to buy materials expensively from a competitor in order to honour marketing contracts).

Transport of materials

The next stage in the laboratory preparation is to carry the reagents to the laboratory bench and mix them in appropriate proportions. On an industrial scale, this may well involve using an electrically driven pump to move each reagent to the reaction vessel through a series of pipes.

In a chemical plant, materials have to be moved between different stages: reactors, distillation columns, and so on. This is particularly important in continuous processes where there must be a steady flow of reagents into the reactor and finished products out of the reactor. Because fluids are easier to meter and to control than solids, most continuous processes involve material in the liquid or gaseous state.

Extension

Transfer of heat

In our laboratory process, heat is supplied by a Bunsen flame. An industrial process may involve thousands of times as much heat. Burning natural gas or other fossil fuels is the usual source of heat.

In the chemical industry, heat is used in several ways, such as

- to raise the temperature of reactants so that a reaction begins or continues;
- to heat mixtures of liquids to separate and purify products by distillation;
- to dry product material;
- to process waste treatment.

The heat used in separation, drying and waste management may be more than that used in the reaction stages.

Heat is expensive and so heat transfer must be carried out as efficiently as possible*.

* We will discuss this further in Unit 48.

Separation

Going back to our laboratory preparation: after crystallization, the ammonium salt is filtered off using a filter paper and funnel. On the industrial scale, this corresponds to the end of the synthesis stage and the beginning of the separation stage.

Most separation techniques used in the laboratory, such as filtration, distillation and solvent extraction, have their industrial equivalents. In addition, chemists have developed other separation techniques which are not suitable for laboratory use.

Process control

In a laboratory we normally rely on our senses to tell us when things are not going according to plan, so that we can quickly decide what adjustments are necessary and carry them out. For instance, our eyes enable us to see when the contents of an evaporating dish are being overheated and our brain responds by instructing our hands to move the dish out of the flame.

With a large-scale continuous process it is neither practical nor desirable to use this kind of manual control, and one of the characteristics of a modern chemical plant is a sophisticated automatic control system.



Fig. 47.16 Technicians checking machinery for the recycling of chemicals

Extension

Sensors are used around the plant to measure variables such as temperature, pressure and flow rate. This information is transmitted to an automatic controller (often an on-line computer) which is programmed to adjust control valves in order to maintain the desired operating conditions and to ensure a consistent quality of product. For maximum efficiency, automatic process control has become increasingly important.

47.8 NPK compound fertilizers

Fig. 47.17 shows the flow plan for the manufacture of NPK fertilizers. Big manufacturers make ammonia, nitric acid, sulphuric acid and phosphoric acid in the same site. They mix the products ammonium phosphate and ammonium nitrate with potassium chloride which is mined. As fertilizer is a bulky product, the cost of handling and transportation is high. Therefore it is desirable to locate production plants near to end users.

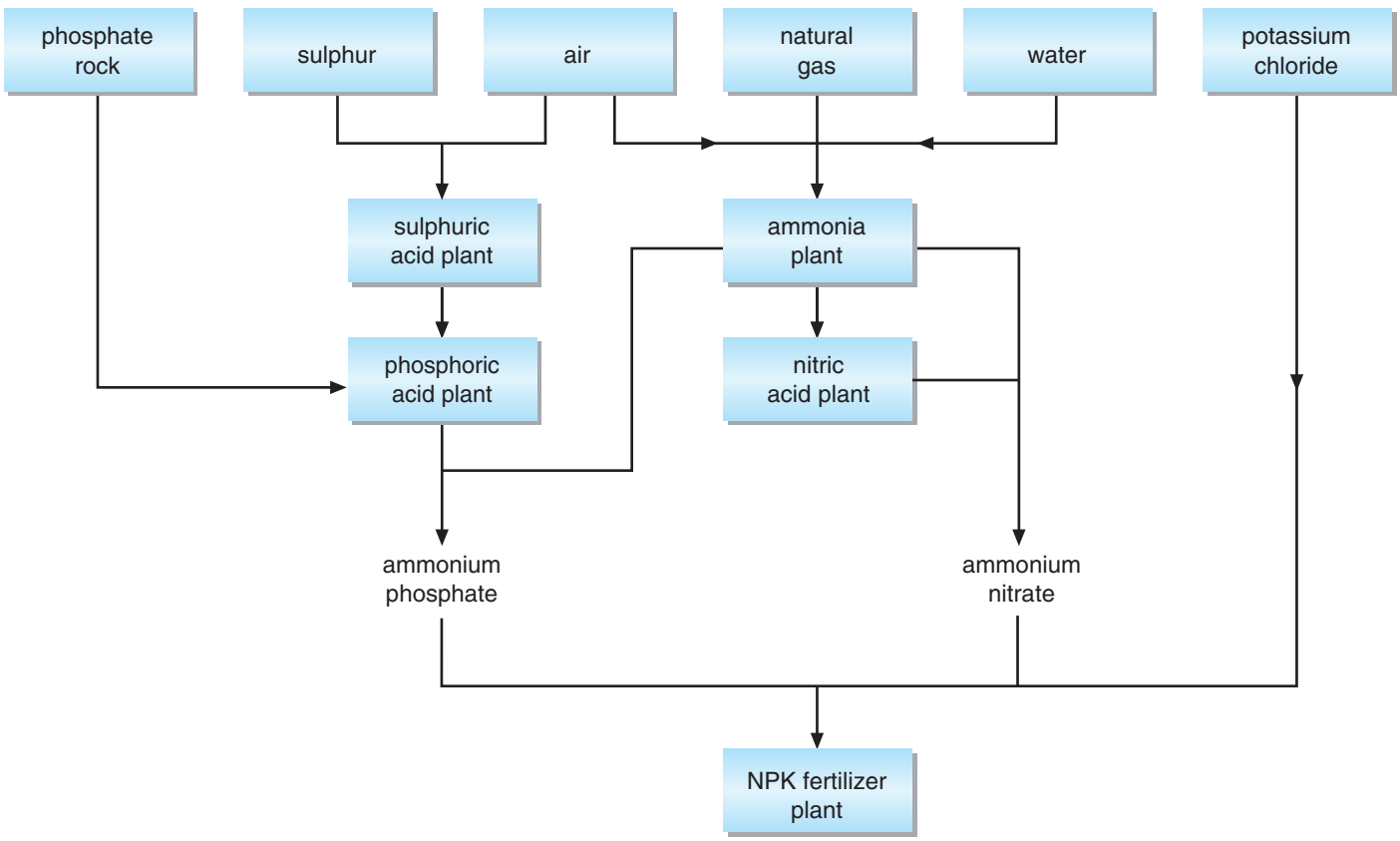


Fig. 47.17 Flow plan for the manufacture of NPK fertilizers

Fertilizers are hygroscopic which readily pick up moisture and may 'cake'. 'Caking' leads to difficulties in the storage, transportation and spreading of the product. Manufacturers overcome this problem by producing fertilizers which are protected, either by external coatings or as a result of their internal compositions.

The main cost element in ammonia production is the price of feedstocks, the most important being natural gas. Therefore measures to improve the production process have focused on reducing the amount of hydrocarbon feedstock required to produce a tonne of ammonia.

47.9 The pros and cons of fertilizers



Fig. 47.18 Tea break on a British farm in the 1930s

A hundred years ago, most farms were small family farms. They used animal manure and kitchen waste as fertilizer. As machinery improved they could farm more efficiently, so food output increased — but slowly.

Then in 1913 the first ammonia plant was set up in Germany, using the Haber process. The production of nitrogenous fertilizers took off after the World War II, and so did crop yields per hectare. Fig. 47.19 shows changes in wheat yield per hectare in Britain since 1948. Some of this increase was due to improved farm machinery and plant breeds, but much was the result of using nitrogenous fertilizers.

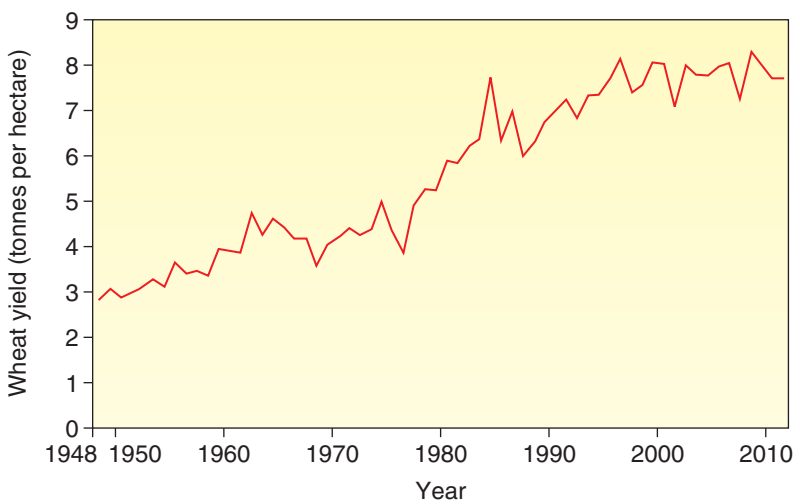


Fig. 47.19 Average wheat yield in the United Kingdom per hectare after World War II (Source: Department for Environment, Food and Rural Affairs, United Kingdom)

Enough food to feed the world?

Look at how the world's population has grown in the last 50 years (Fig. 47.20). How has it been fed? The land available for farming has not changed a great deal. What has changed most is the use of nitrogenous fertilizers.

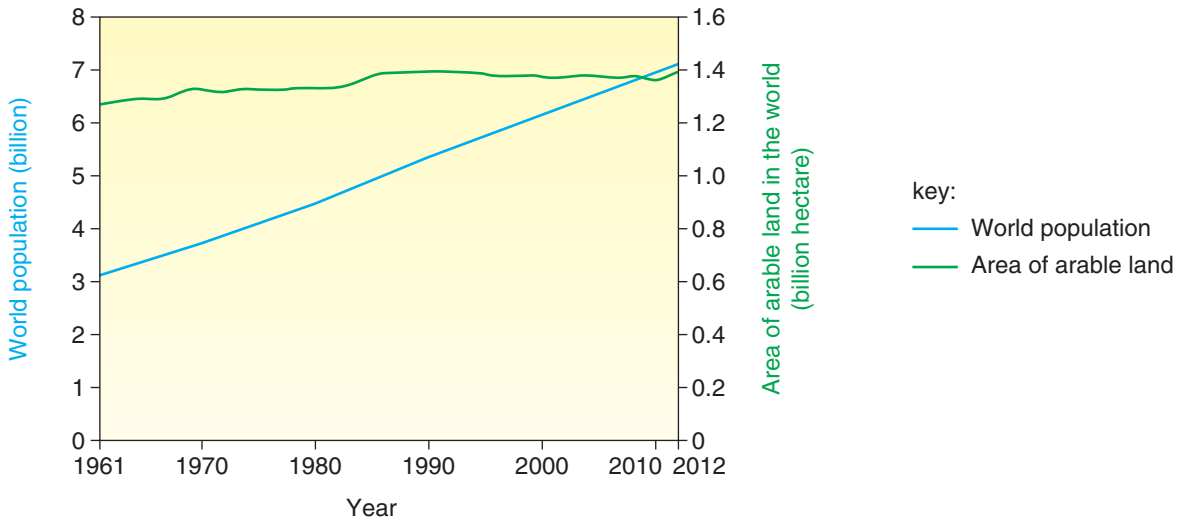


Fig. 47.20 World population shown against world trend in area of arable land (Source: FAOSTAT)

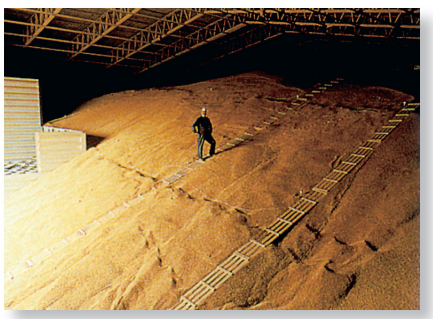


Fig. 47.21 Part of the European 'grain mountain'. Is it right that huge amounts of food are kept in storage while many people are starving?

It is estimated that 40% of the world's over 7 billion people are alive today due to the Haber process.

Today the world produces more than enough food to feed everyone. But many people are still starving. This is because food is not shared around the world equally. Some poor countries produce too little food because farmers cannot afford fertilizers, or better seeds, or machinery, and rain often fails. Other countries produce — and consume — more than they need.

By 2050, there could be 9.2 billion people to feed. Without fertilizers this would be impossible.

Agriculture — the key to poverty reduction

Agriculture plays a central role in the well-being of developing countries' economies and their people. In the developing world, an average of 50% of people make their living from farming and agriculture, and in some countries this figure rises to over 80%.



Fig. 47.22 Agriculture is important to developing countries

Three quarters of the 1.2 billion people who earn less than one US dollar a day work and live in rural areas. Studies show that agriculture is the key in the fight against poverty. Nitrogenous fertilizers help reduce poverty by increasing food availability, reducing food prices and increasing rural incomes.

Environmental problems associated with the use of fertilizers

When a fertilizer is sprayed on the soil, it does not go just to plant roots. Fig. 47.23 shows the fate of nitrogenous fertilizers when added to cereal crops.

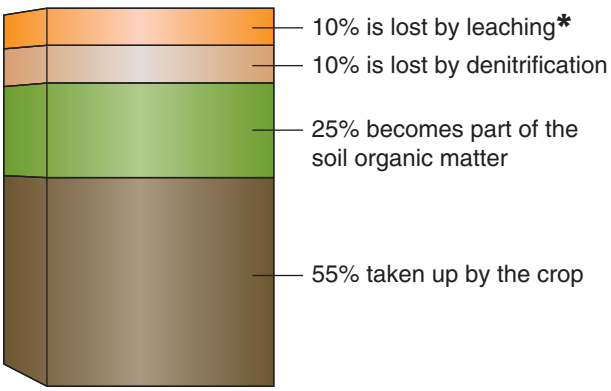


Fig. 47.23 The fate of nitrogenous fertilizers when added to cereal crops

* *Leaching* refers to the loss of water-soluble plant nutrients from the soil, due to rain and irrigation.

Fig. 47.24 shows environmental problems associated with the use of fertilizers.

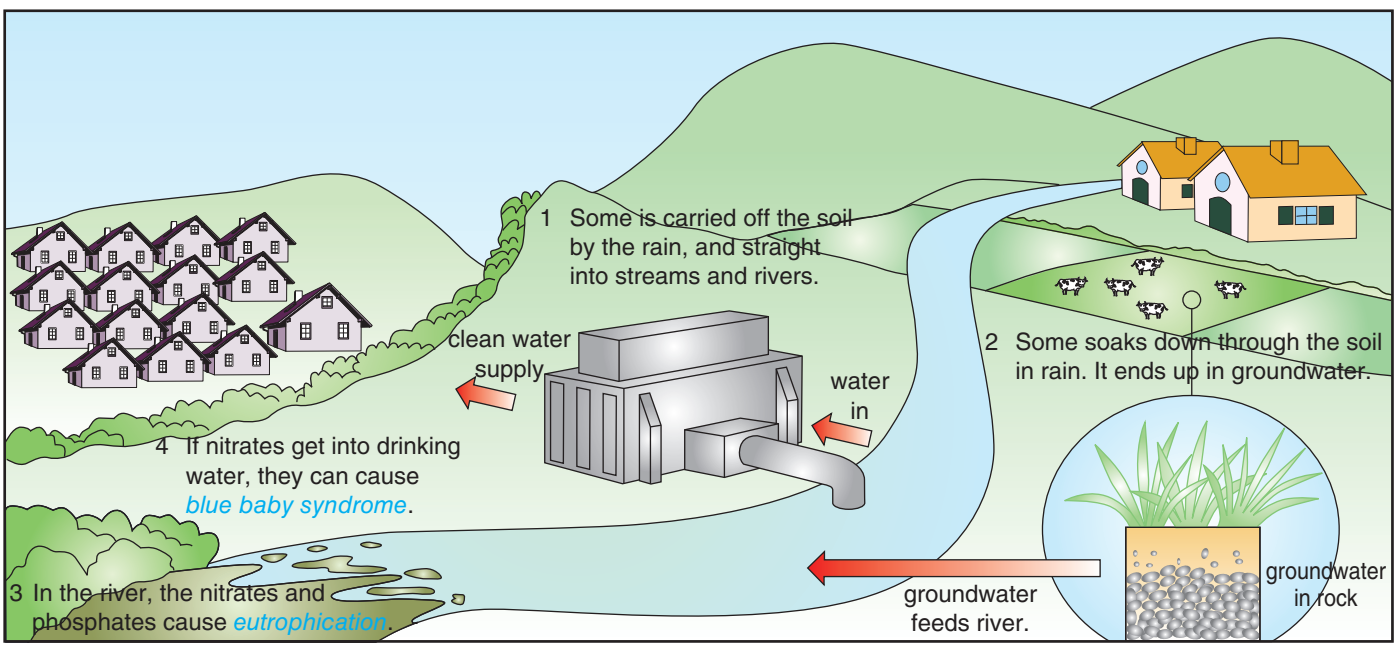


Fig. 47.24 Environmental problems associated with the use of fertilizers

leaching 瀝濾　　eutrophication 富養化　　blue baby syndrome 藍嬰綜合症

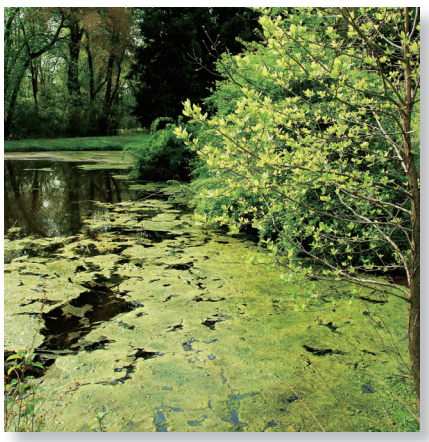


Fig. 47.25 The surface of this lake is covered with algae

Animation
Eutrophication

Eutrophication

Leaching of fertilizers from the ground by water causes nitrates and phosphates to enter rivers. Nitrates and, to a greater extent, phosphates help the growth of algae.

Algae float on the river like a carpet and cut off sunlight from the plants below. Consequently, these underwater plants cannot photosynthesize and they die. Bacteria feed on the decaying plant material. The bacteria's respiration uses up most of the dissolved oxygen. As a result, the fish die. The whole process — which results in the death of a river — is known as eutrophication.

Blue baby syndrome

Bacteria in our intestine may convert nitrates to nitrites. Nitrites combine with haemoglobin, thus reducing the oxygen-carrying capacity of blood. This can cause oxygen starvation in young infants. Their skins take on a blue tinge. The development of infants may be affected. This is known as the blue baby syndrome.

A high concentration of nitrates in drinking water has also been linked to cancers, but this link has not been proven.

47.10 The chloroalkali industry

AL 2008 Paper 1 Q9

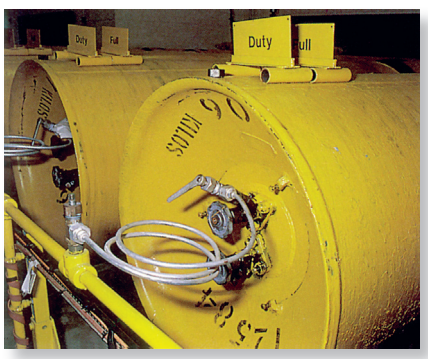


Fig. 47.26 Tanks for chlorine at a waterworks. What is the use of chlorine here?

A whole section of the chemical industry has developed around a single electrolysis process. It is called the **chloroalkali industry**, and it is based on the electrolysis of concentrated sodium chloride solution. The electrolysis process gives three important industrial chemicals — chlorine gas, hydrogen gas and sodium hydroxide solution.

Chlorine is one of the most important chemicals in the chemical industry. It is used as a disinfectant, in water treatment and in processing pulp and paper. Large amounts of chlorine are needed for polyvinyl chloride, an important material in the building industry as well as for consumer goods.

Fig. 47.27 shows the common uses of hydrogen, chlorine and sodium hydroxide*.

* We have discussed this in Topic 1 Planet Earth.

chloroalkali industry 氯鹼工業

CE 2007 Paper 2 Q41

CE 2004 Paper 2 Q36
AL 2009 Paper 1 Q3(a)(iv)

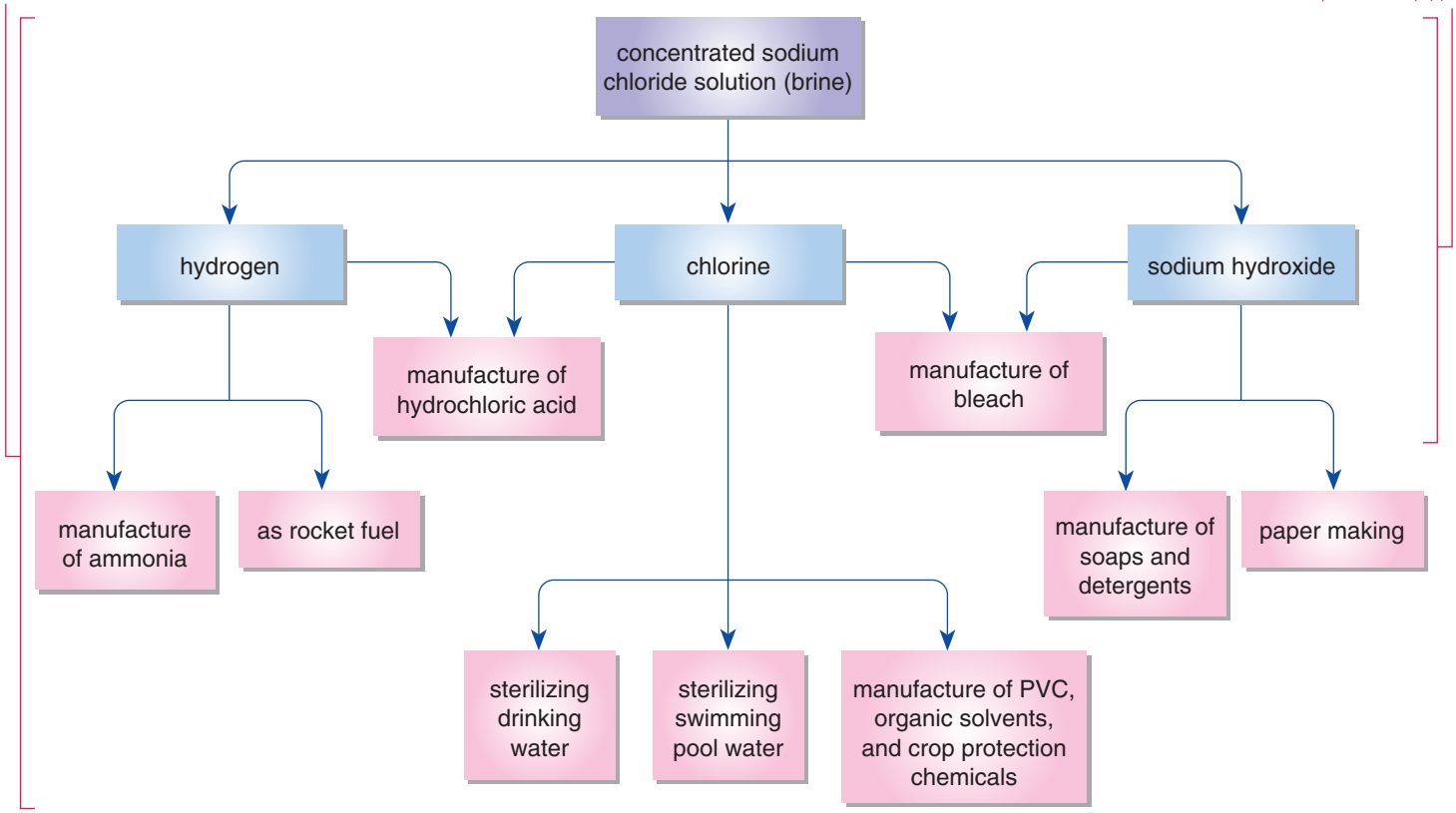


Fig. 47.27 Common uses of hydrogen, chlorine and sodium hydroxide

47.11 Manufacture of chlorine

Electrolysis of brine

DSE 2014 Paper 2 Q1(a)(iv)
DSE 2015 Paper 2 Q1(c)(i)

We can obtain hydrogen, chlorine and sodium hydroxide solution from the electrolysis of concentrated sodium chloride solution (brine).

Two problems occur if brine is electrolyzed in a simple container:

- Chlorine and hydrogen gases may mix together. They may react to form hydrogen chloride gas.
- Chlorine may react with the sodium hydroxide solution directly in the container.

This means that the products must be kept apart throughout the process. Three types of cells have been developed for the purpose.

Mercury electrolytic cell



Fig. 47.28 An industrial mercury electrolytic cell

In the mercury electrolytic cell (Fig. 47.29), the cathode is a shallow layer of mercury. During electrolysis, chlorine gas is produced at the carbon anodes and sodium is produced at the mercury cathode.

DSE 2013 Paper 2 Q1(c)(i)

At the anode

$$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$$

At the cathode

$$Na^+(aq) + e^- + Hg(l) \longrightarrow \underset{\text{sodium amalgam}}{Na/Hg(l)}$$

The sodium dissolves in mercury to form an **amalgam**. The amalgam then flows into a second cell and reacts with water to form hydrogen gas and sodium hydroxide solution. Mercury is recovered.

DSE 2013 Paper 2 Q1(c)(i)

$$\underset{\text{sodium amalgam}}{2Na/Hg(l)} + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g) + 2Hg(l)$$

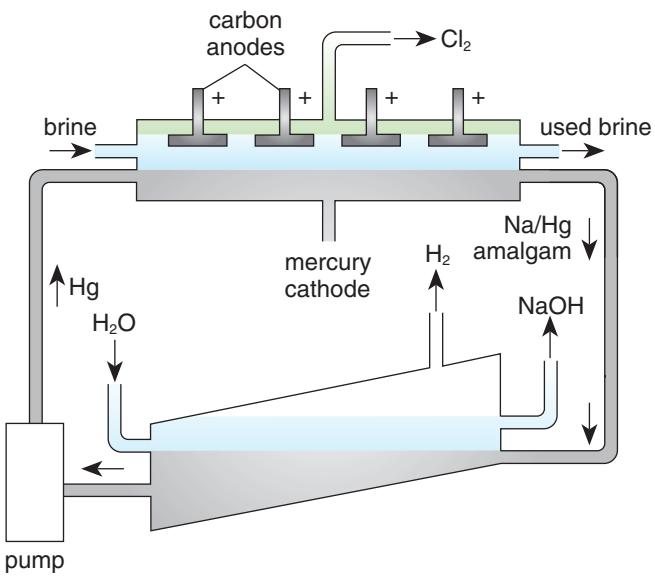


Fig. 47.29 Mercury electrolytic cell

This process is very efficient and gives products of very high purity.

Unfortunately, this process also produces waste which contains poisonous mercury compounds. These waste products cause serious pollution problems if they are discharged into rivers and seas. The marine animals may die of mercury poisoning. N6

DSE 2015 Paper 2 Q1(c)(iv) AL 2008 Paper 1 Q9

Animation
Manufacture of chlorine by electrolysis of brine — Mercury electrolytic cell

Reference Website
Refer to the following website for an animation showing the operation of a mercury electrolytic cell:
http://www.eurochlor.org/the-chlorine-universe/how-is-chlorine-produced/the-mercury-cell-process/mercury-cell-process.aspx
(accessed May 2015)

N6 Mercury pollution
Mercury is extremely insoluble. If it stayed as such at the bottom of the ocean, it would be of little concern. Unfortunately there are some bacteria that convert mercury into compounds such as dimethyl mercury, $Hg(CH_3)_2$. Compounds of mercury in ocean water can be taken up by organisms and passes along the food chain. The result is that end consumers, such as human beings who eat these fish (particularly tuna and shark), can receive quite significant doses of mercury. Mercury affects the nervous system and can cause brain damage; in addition, it tends to stay within the human body for long times.

Consequently environmental protection agencies in many countries have set limits on the amount of mercury that chloroalkali plants can discharge. Typical figures of less than 1 g per tonne of NaOH produced are very difficult to meet. Hence very few new mercury electrolytic cells are now being built.

amalgam 汞齊

Extension

Diaphragm cell

In the diaphragm cell, a diaphragm separates the anode and cathode compartments, and does not allow gases to pass through. Brine enters the anode compartment where chloride ions are discharged to form chlorine gas. The resulting solution, rich in sodium ions and also containing some chloride ions, seeps through the diaphragm into the cathode compartment. Here hydrogen ions are discharged to form hydrogen gas. A solution containing both sodium hydroxide and unreacted sodium chloride is formed in the cathode compartment.

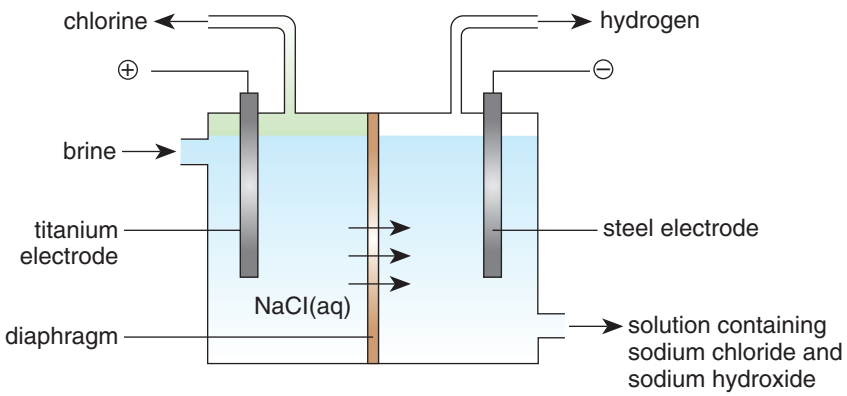


Fig. 47.30 A diaphragm cell

AL 2008 Paper 1 Q9

Membrane cell N7

CE 2010 Paper 1 Q11(f)

In many countries, mercury cells and diaphragm cells have been replaced by membrane cells. In the membrane cell, the anode and cathode compartments are separated by a water-impermeable plastic membrane. Brine flows through the anode compartment and water flows through the cathode compartment. The membrane allows sodium ions to pass but not hydroxide or chloride ions. Thus, pure sodium hydroxide and hydrogen gas are produced at the cathode and chlorine gas is evolved at the anode. N8

DSE 2013 Paper 2 Q1(c)(i)

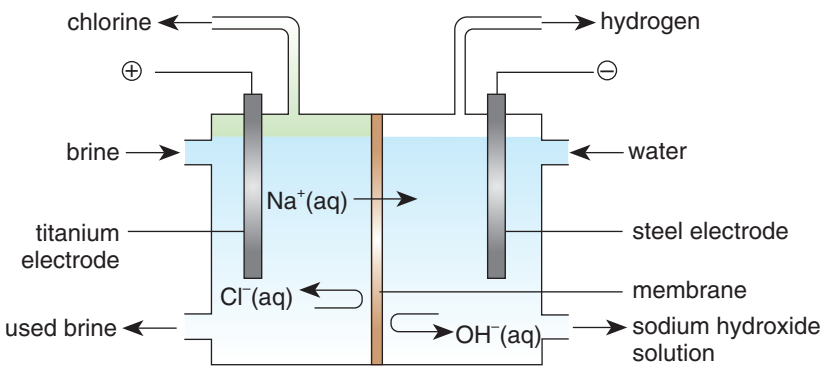


Fig. 47.31 A membrane cell

N7 Examination questions often ask about the membrane cell.

N8 Reactions that occur in a membrane cell

At the anode

$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$

At the cathode

$2H^+(aq) + 2e^- \longrightarrow H_2(g)$

Water dissociates continuously to replace the hydrogen ions discharged at the cathode. Thus, there is a build-up of hydroxide ions near to the cathode.

Some initial purification of salt is required for the membrane cell. Some energy is needed to increase the concentration of the sodium hydroxide for subsequent commercial uses.

Reference Website

Refer to the following website for an animation showing the operation of a membrane cell:
http://www.absorblearning.com/media/item.action?quick=uh
(accessed May 2015)

N9 Refer to page T46.

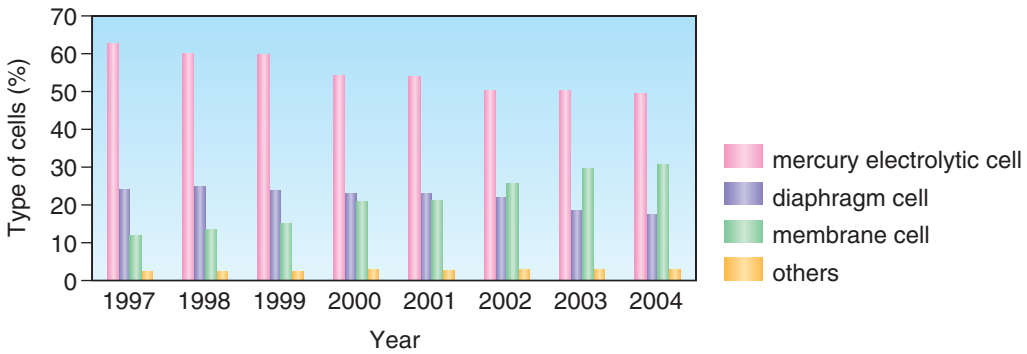


Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Both diaphragm and membrane cells do not use mercury. Hence the risk that mercury poses to health and the environment is eliminated. N9

The diagram below shows the split between various processes used in chlorine production in Europe (1997 – 2004). Taking the case of Europe as an example, discuss why so many plants have not switched to the membrane cell in chlorine production.



Discussion
When production is on an industrial scale, there are many issues to consider and if the present system is working well, why change it? Issues such as cost effectiveness, purity of the product, environmental impact and the disposal of the mercury if a different type of cell is to be employed, all need to be considered. It may cause more environmental damage to change a system than to keep the present system well maintained.
However, when setting up a new processing plant, the environmental concerns are much higher up the agenda today than they were at the start of the 20th century. The membrane cell will be the favourite in the majority of cases.

N10 Methanol is an important compound in the chemical industry. It is a 1-carbon compound and acts as a starting material to make organic compounds with greater carbon numbers.

47.12 Uses of methanol

From recyclable plastic bottles to pharmaceuticals, polyester and paint, methanol is in everybody's life.

Methanol is used as a feedstock in the production of chemicals N10 such as acetic acid (ethanoic acid) and formaldehyde (methanal). It is also used as a feedstock in products like adhesives, plywood subfloors, solvents and windshield washer fluid.

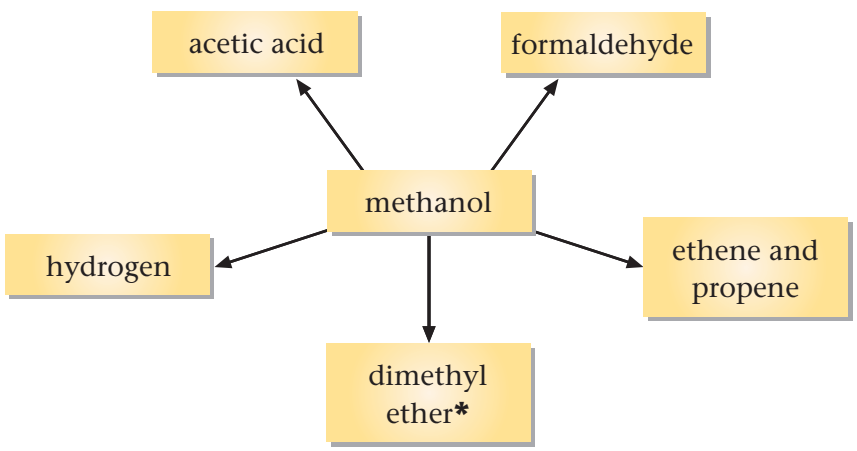
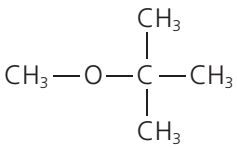


Fig. 47.32 Chemicals produced from methanol

* Dimethyl ether has the following structure:
$CH_3—O—CH_3$

* We have discussed methanol fuel cells in Topic 5 Redox Reactions, Chemical Cells and Electrolysis.

* Methyl tertiary butyl ether has the following structure:



* Fuel oxygenates, such as MTBE, are chemicals containing oxygen that are added to fuels, especially petrol, to make them burn more efficiently, thus reducing air pollution associated with car emissions.

Methanol is a convenient fuel for fuel cells* being developed for mobile phones, laptops and small scale transportation applications. Methanol is also used to make MTBE (methyl tertiary butyl ether)*, an oxygenate* used in petrol.



Fig. 47.33 MTBE notice on a petrol pump



Fig. 47.34 This race car is fuelled by methanol



Fig. 47.35 Gases from oilfields are often disposed of by burning

Besides ethanol, methanol can also be used as a fuel. It burns N11 cleanly in a car engine to form the harmless combustion products carbon dioxide and water, with little carbon monoxide emission. Methanol produces 60% of the energy of an equal amount of petrol. So cars need larger tanks when methanol is used as a fuel.

47.13 From natural gas to methanol

Natural gas represents a global energy resource similar in size to that of petroleum. However, considerable volumes of natural gas are located in remote areas lacking pipelines or other infrastructure to transport them directly to the customer. Substantial amounts are burnt off as an unwanted by-product of petroleum production (Fig. 47.35).

Manufacturers can use Gas-to-Liquids (GTL) processes to convert natural gas to liquid products, primarily liquid hydrocarbons, dimethyl ether or methanol. All of these liquid products can be readily transported by land and sea.

N11 Methanol as a fuel in cars
A few years ago, only a small amount of methanol was used directly as a fuel in cars. This use is now increasing rapidly. In China, petrol blended with 15% of methanol is used as a fuel for vehicles. Modification to the engines is not necessary with this percentage of methanol. With some redesigning, blended fuel with more methanol (up to 85%) can be used.

While the technology for producing each of these products has been developed years ago, only methanol is currently in widespread commercial production.

Currently, about 90% of the worldwide production of methanol is derived from natural gas (the main component being methane).

Today's methods of making methanol have three main stages:

- *syngas* (synthesis gas) generation;
- methanol synthesis;
- crude methanol purification.

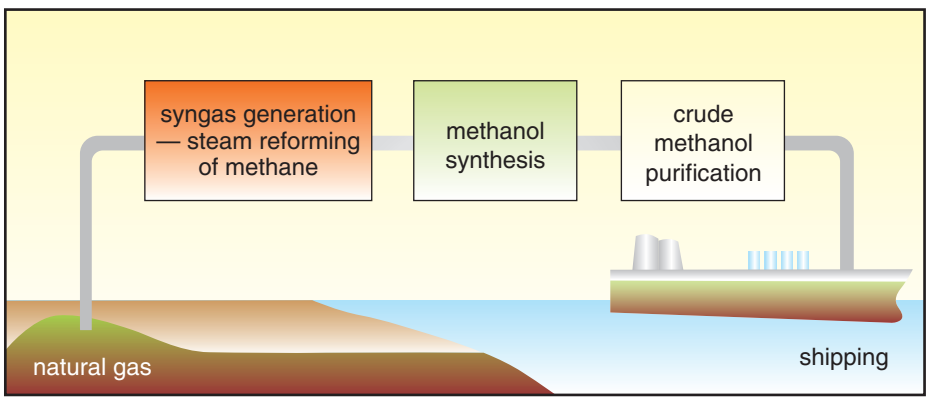


Fig. 47.36 Methanol manufacturing has three main stages

The first stage of the production process is converting the methane into syngas, a mixture of primarily carbon monoxide and hydrogen — the steam reforming of methane. After the removal of sulphur impurities, methane and steam react over a nickel oxide catalyst at 730 °C and 30 atmospheres according to the following equation:

$$CH_4(g) + H_2O(g) \xrightleftharpoons[730\ °C,\ 30\ \text{atmospheres}]{\text{nickel oxide catalyst}} CO(g) + 3H_2(g)$$

N12 Students need to remember the reaction conditions and chemical equations for the
- synthesis of ammonia in the Haber process;
- manufacture of nitric acid from ammonia;
- syngas generation from methane;
- synthesis of methanol from syngas.

Methanol synthesis is carried out over a copper and zinc oxide catalyst at around 300 °C and 50 – 100 atmospheres according to the following equation: N12

$$CO(g) + 2H_2(g) \xrightleftharpoons[300\ °C,\ 50 - 100\ \text{atmospheres}]{\text{copper and zinc oxide catalyst}} CH_3OH(g) \qquad \Delta H = -129\ \text{kJ mol}^{-1}$$

DSE 2012 Paper 2 Q1(b)(ii)

syngas 合成氣

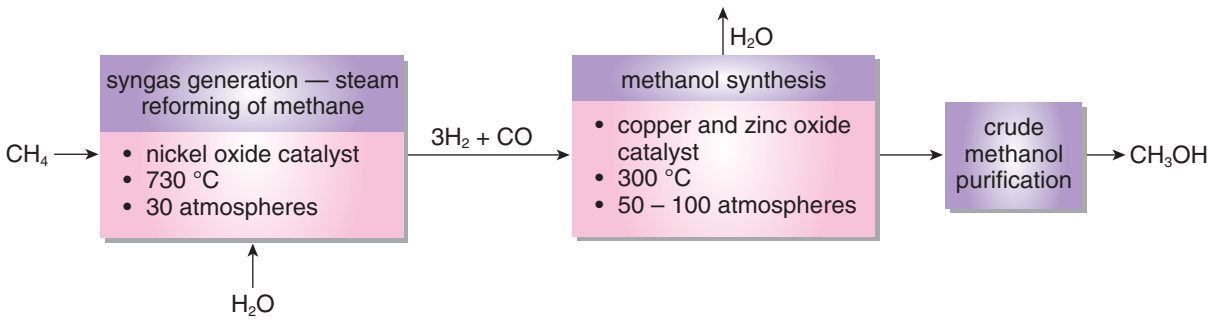


Fig. 47.37 A flow diagram for methanol synthesis from natural gas

It is actually thought that most of the methanol is produced from carbon dioxide formed from the reaction of carbon monoxide with steam.

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

$$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g)$$

The crude methanol produced contains both water and low levels of by-products. Pure methanol is obtained by fractional distillation.

Changes in the production process of methanol

Virtually any hydrocarbon source (coal, petroleum, naphtha, etc.) can be converted to methanol via syngas. The production of methanol from syngas on an industrial scale using a high pressure (250 – 350 atmospheres) and temperature (300 – 400 °C) was first introduced during the 1920s. From then until the end of World War II, methanol was mainly produced from coal-derived syngas which contained a lot of impurities.

After World War II, the raw material for methanol synthesis shifted rapidly from coal to natural gas. Syngas derived from natural gas had fewer impurities. This allowed the use of more active catalysts operating under milder conditions. During the 1960s, a process using a low pressure (50 – 100 atmospheres) and temperature (200 – 300 °C) was developed. The main advantages of the low-pressure process are lower production costs, improved operational reliability and greater flexibility in the choice of plant size. The low-pressure process has now replaced the high-pressure process.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Refer to page T47.

The following reaction is involved in the synthesis of methanol:

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \qquad \Delta H < 0$$

The conversion is carried out at around 300 °C and a pressure of 100 atmospheres. About 10% conversion to methanol is obtained.

a) Predict what will happen to the percentage of methanol in the equilibrium mixture if

i) the temperature is increased;

ii) the pressure is increased.

b) Explain the reasons for the choice of reaction conditions in terms of achieving

i) a high yield;

ii) a high rate.

47.14 Methanol — a green feedstock?

Syngas for methanol synthesis can be produced from almost any kind of hydrocarbon, such as natural gas, coal and even *biomass*. Methanol can be a green feedstock* as well as a green fuel for cars if we can produce syngas from renewable resources for methanol synthesis.

DSE 2012 Paper 2 Q1(b)(iii)

* We will discuss green chemistry further in Unit 48.

Methanol from biomass

Biomass is organic matter such as urban wood waste, agricultural residues and animal waste that can be made into fuel.



Fig. 47.38 Making fuel blocks from cow dung in India

biomass 生物量

* Gasification is a reaction that occurs when organic matter is heated to high temperatures without much oxygen.

Fig. 47.39 shows a flow diagram for the synthesis of methanol from biomass, including hydrogen generation by electrolysis. The biomass is converted to a product gas in a process called **gasification***. The product gas contains hydrogen, carbon monoxide, carbon dioxide, etc.

DSE 2012 Paper 2 Q1(b)(iii)

Extension

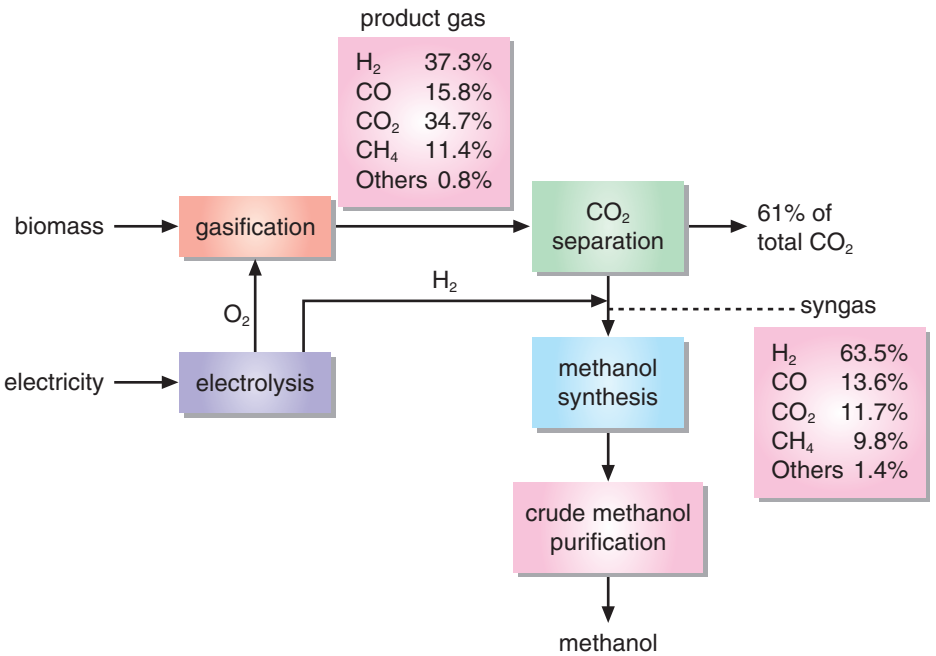


Fig. 47.39 A flow diagram for the synthesis of methanol from biomass, including hydrogen generation by electrolysis

The gasification of biomass does not produce a syngas with the quality required for methanol synthesis. The product gas contains too much carbon dioxide. Therefore part of the carbon dioxide is removed from the raw gas before methanol synthesis. The addition of hydrogen gives a syngas that is suitable for methanol synthesis.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

Places with lots of biomass and renewable energy sources for generating electricity (e.g. hydroelectric power) are ideal for applying this process.

gasification 氣化

Methanol from carbon dioxide captured from flue gas

Methanol can be made by combining carbon dioxide and hydrogen:

$$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g)$$

Fig. 47.40 shows a flow diagram for the synthesis of methanol from carbon dioxide. The carbon dioxide can be captured from flue gas. In such a system, the carbon dioxide released by burning methanol can be cancelled out by the carbon dioxide captured to make it.

DSE 2012 Paper 2 Q1(b)(iii)

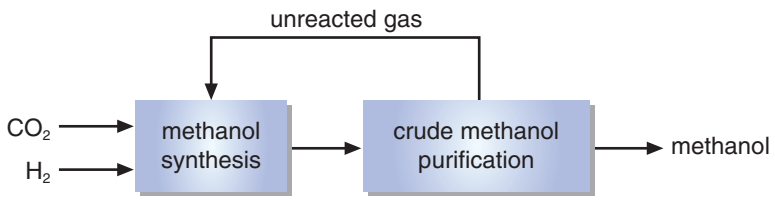


Fig. 47.40 A flow diagram for the synthesis of methanol from carbon dioxide

The process requires considerable energy to generate hydrogen from water by electrolysis, but that may come from renewable energy sources.

Extension

47.15 Choosing a site for a chemical plant

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



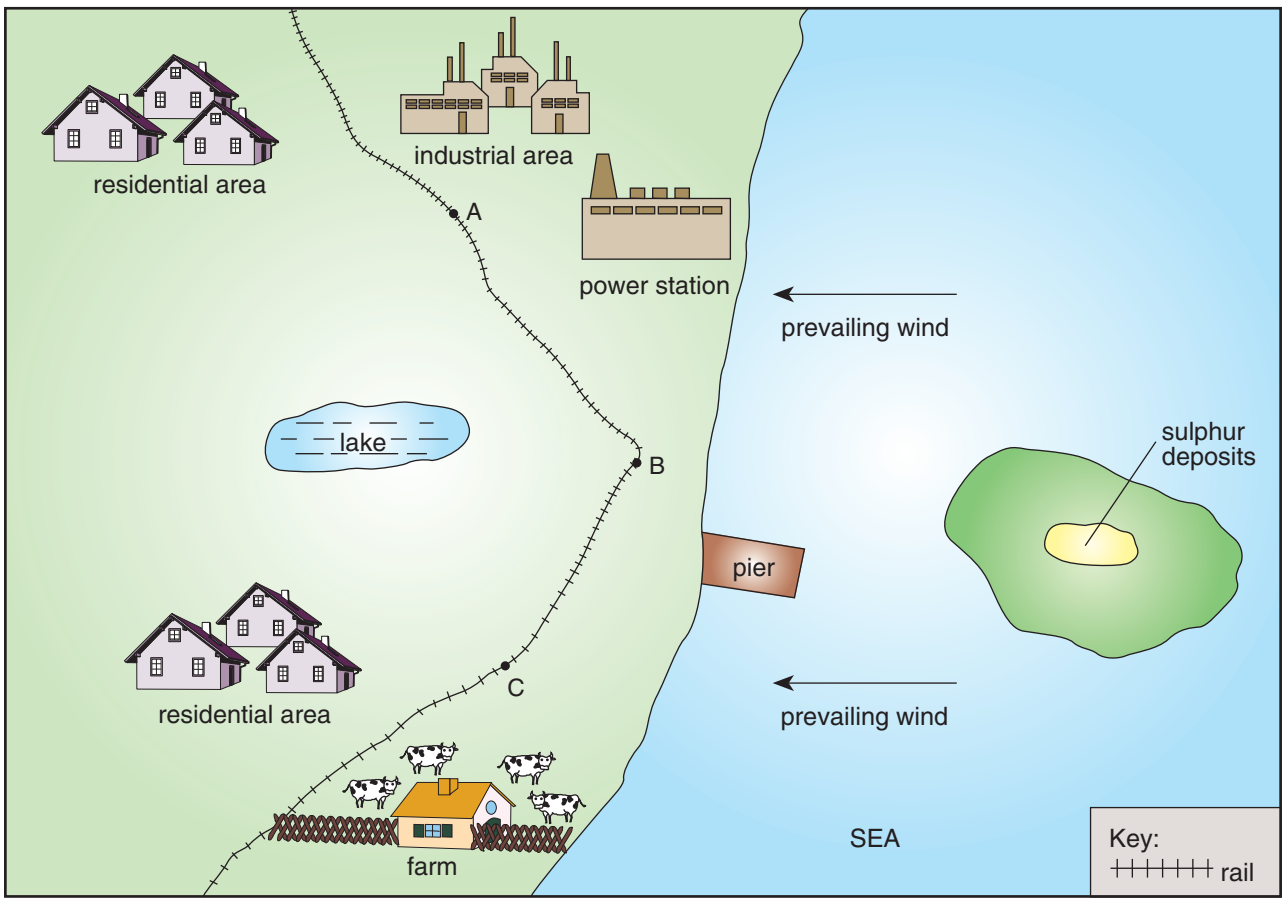
Decision Making

Refer to page T48.

In Unit 45, we have considered the case of a chemist working on the feasibility of manufacturing a new chemical in industrial scale. Suppose his / her study indicates that the manufacture is feasible and profitable. The company decides to build a new chemical plant for that. The first thing is choosing a site for the new chemical plant.

Suppose you were the director of this company. Your company plans to build the new plant on the mainland of a city as shown below. Three possible locations (A, B and C) are identified.

Extension



Remark:

- The company has recently discovered sulphur deposits on an outlying island near to the mainland. Sulphur is a feedstock for the production.
- Sea water is not suitable for the production process.

1 Make a list of criteria you would use to assess the suitability of a site.

2 Decide where you would build the plant. Explain your choice based on the list made above.

ASL 2003 Q14

To choose a site for a new chemical plant, the following factors may be considered.

1 **Easy availability of raw materials** If a plant is close to where raw materials are produced or mined, transportation cost will be reduced.

2 **Reliable supply of electricity and fuel** This is important in keeping the production going.

Extension

3 **Reliable supply of water** Water is important to chemical processes, not only as a raw material but also for cooling, cleaning, etc.

4 **Good transport system** A plant should be close to a cheap, efficient road transport system for delivery of the product to consumers. If the product is for overseas markets, a deep-water port may be more important.

5 **Good supply of labour** The plant should be close to areas where labour is available.

6 **Cheap land resources** This is important if the plant is large in scale.

7 **Least environmental impact** Chemical plants are likely to produce chemical waste and cause pollution problems. Such plants should be located at sites where the environment is least likely to be affected. A plant close to waste treatment facilities is desirable.

ASL 2003 Q14

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

VOCs and photochemical smog

VOCs mean volatile organic compounds. Many VOCs are hydrocarbons. Paints, the printing industry, various related consumer products (such as aerosol sprays) and vehicles are the four major emission sources of VOCs in Hong Kong.

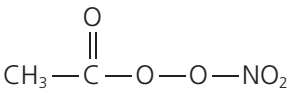


Fig. 47.41 VOCs are found in a wide variety of products such as solvent-based paints, printing inks and many consumer products

How do VOCs cause photochemical smog?

VOCs are major contributors to photochemical smog formation. The word 'smog' comes from two words: smoke and fog. It is used to characterize a visible combination of smoke and fog.

* The structure of peroxyacetyl nitrate (PAN) is shown below:



Photochemical smog is a mixture of pollutants formed by the reaction of nitrogen oxides and VOCs under the influence of sunlight. Pollutants in photochemical smog include particulates, nitrogen oxides, ozone, aldehydes, peroxyacetyl nitrate (PAN)* and hydrocarbons, etc.



Fig. 47.42 Photochemical smog caused by the reaction of pollutants under sunlight

The chemistry of photochemical smog is quite complex. In the initial step, nitrogen dioxide molecules absorb sunlight and break apart to form nitrogen monoxide molecules and free oxygen atoms which are very reactive.

The initiating step $NO_2(g) \xrightarrow{\text{sunlight}} NO(g) + O(g)$

The oxygen atoms then react with hydrocarbons (principally VOCs) and other components of the atmosphere (e.g. oxygen) to produce a variety of irritating and toxic chemicals.

$O(g) + O_2(g) \longrightarrow O_3(g)$

$O(g) + \text{hydrocarbons} \longrightarrow \text{aldehydes}$

$O_3(g) + \text{hydrocarbons} \longrightarrow \text{aldehydes}$

$\text{hydrocarbons} + O_2(g) + NO_2(g) \longrightarrow \text{peroxyacetyl nitrate}$

ASL 2006 Q9(b)

How does photochemical smog affect us?

ASL 1999 Q10(f)(ii)

Photochemical smog can irritate our eyes, nose and throat, or can worsen existing heart and respiratory problems such as *asthma*. People with heart and lung problems, the elderly and children are most at risk.

asthma 哮喘

What has the HKSAR Government done to reduce emissions of VOCs?

The HKSAR Government has implemented measures to control emissions of VOCs from motor vehicles, power plants and factories.

In April 2002, in an effort to address the photochemical smog problem, the HKSAR Government and the Guangdong Provincial Government reached a consensus to reduce the emissions of VOCs by 55% by 2010, using 1997 as the base year.

Questions Refer to page T48.

1 Fig. 47.43 shows how concentrations of substances associated with photochemical smog vary over a 24-hour period in a certain city.

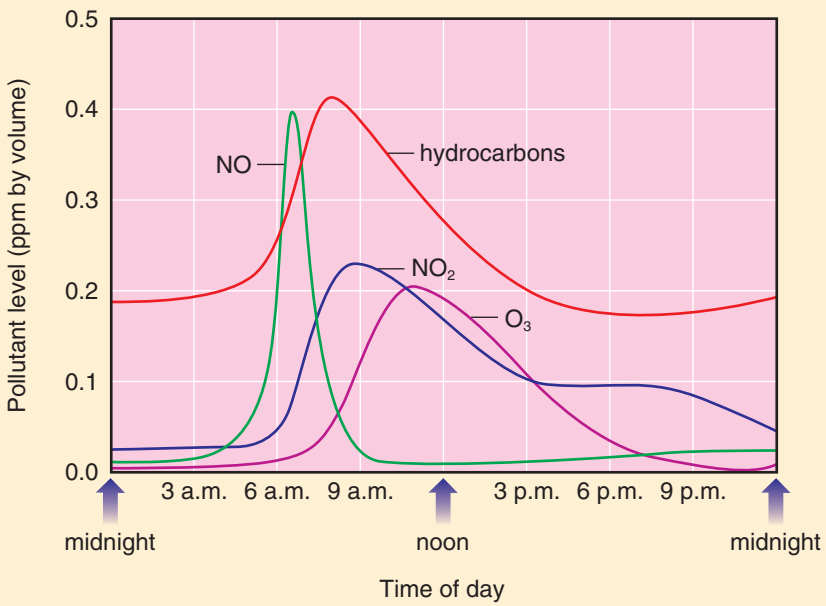


Fig. 47.43 Pollutant level at different times of the day

a) i) Between what hours do concentrations of nitrogen oxides and hydrocarbons peak?

ii) Account for this fact in terms of traffic patterns.

b) Ozone is a major component of photochemical smog.

i) Which substance is at a minimum concentration when ozone is at maximum concentration?

ii) What does this suggest about the production of ozone in polluted air?

2 Suggest how you can help reduce emission of VOCs to the air.

Summary

1 There are three main ways of fixing nitrogen from the air:

a) a chemical reaction in the air during lightning flashes;

b) the action of bacteria;

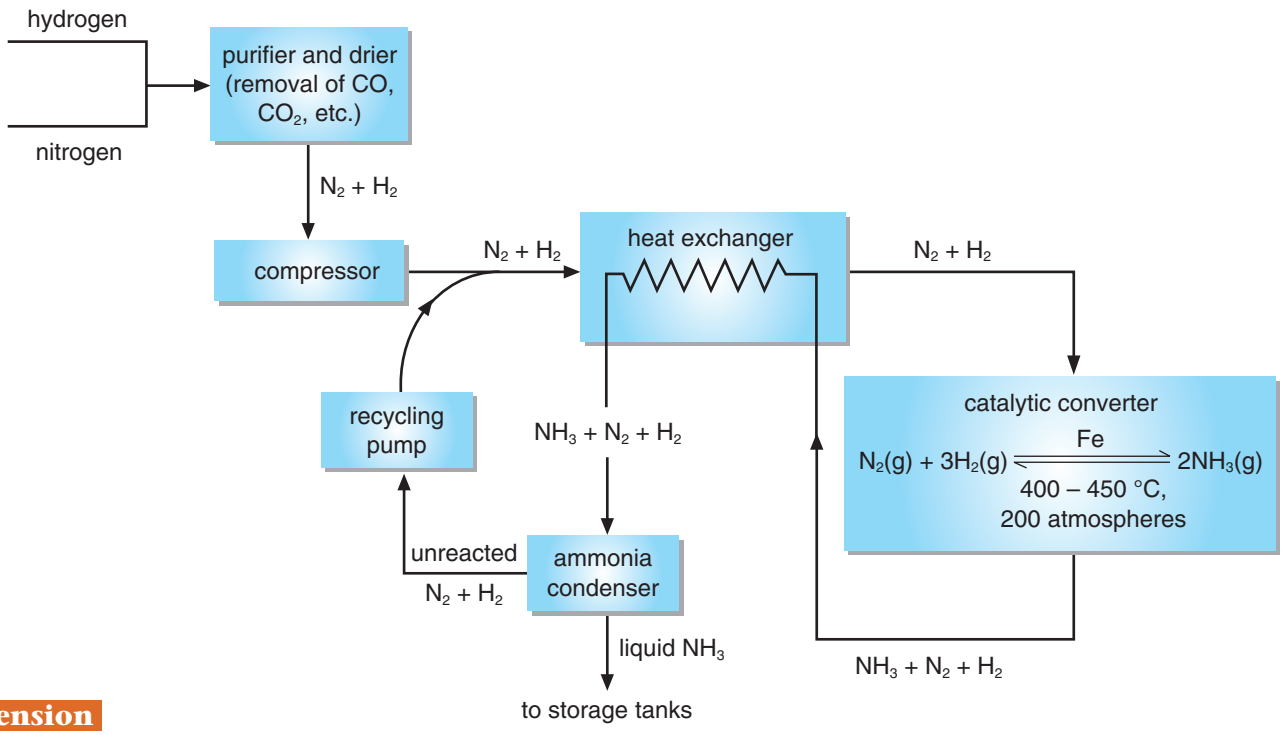
c) the manufacture of ammonia by the Haber process.

2 There are two main types of nitrogenous fertilizers:

a) straight N fertilizers for farmers who want only nitrogen (N);

b) NPK compound fertilizers for farmers who want nitrogen (N), phosphorus (P), and potassium (K).

3 Ammonia is the key to nitrogenous fertilizers. The following flow diagram outlines the Haber process for its synthesis.



Extension

4 When working on large scale production of nitrogenous fertilizer, it is necessary to consider the following issues:

a) storage of raw materials;

b) transport of materials;

c) transfer of heat;

Extension

d) separation;

e) process control.

5 The chloroalkali industry is based on the electrolysis of concentrated sodium chloride solution. The electrolysis process gives three important industrial chemicals — chlorine gas, hydrogen gas and sodium hydroxide solution.

6 The three main types of cell used for the electrolysis of brine are:

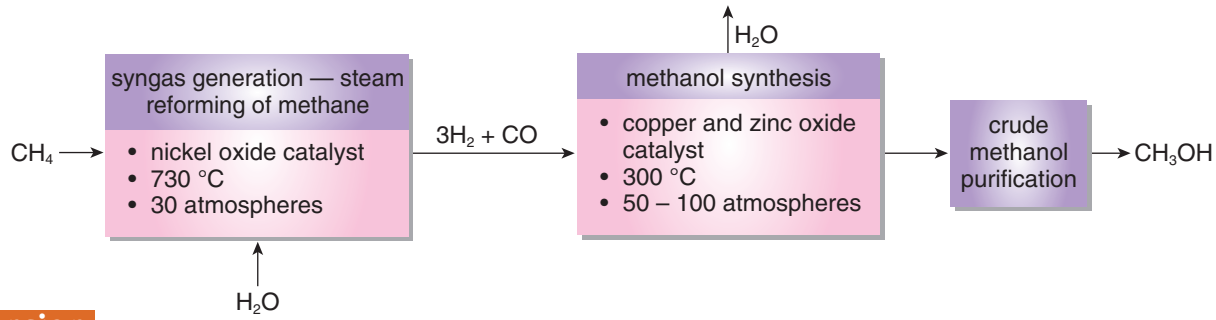
a) mercury electrolytic cell;

Extension

b) diaphragm cell;

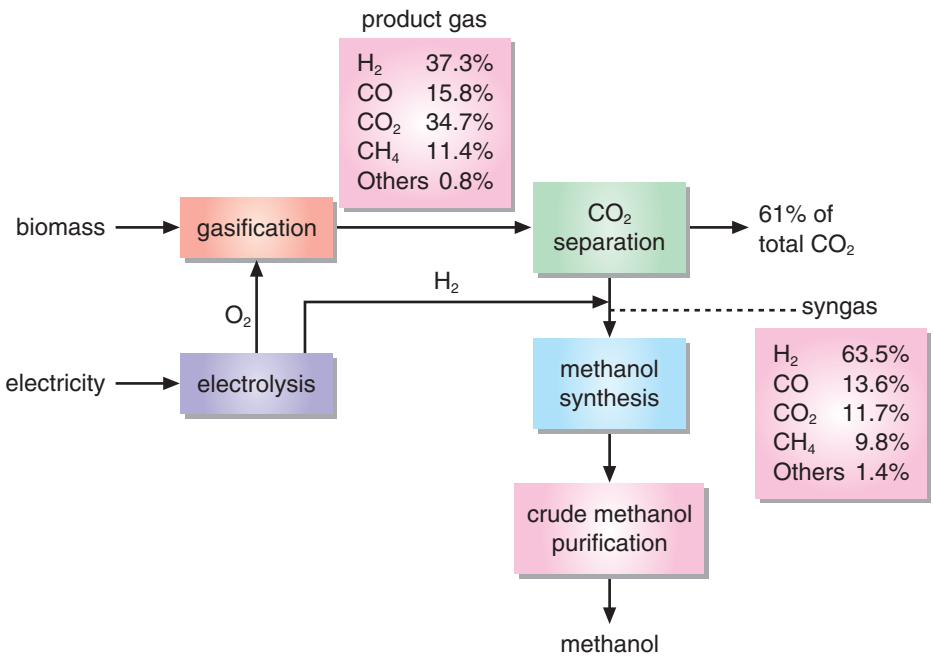
c) membrane cell.

7 Methanol is used as a feedstock in the production of many chemicals. It is produced mainly from natural gas as shown below:



Extension

8 The following diagram shows the synthesis of methanol from biomass:



Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

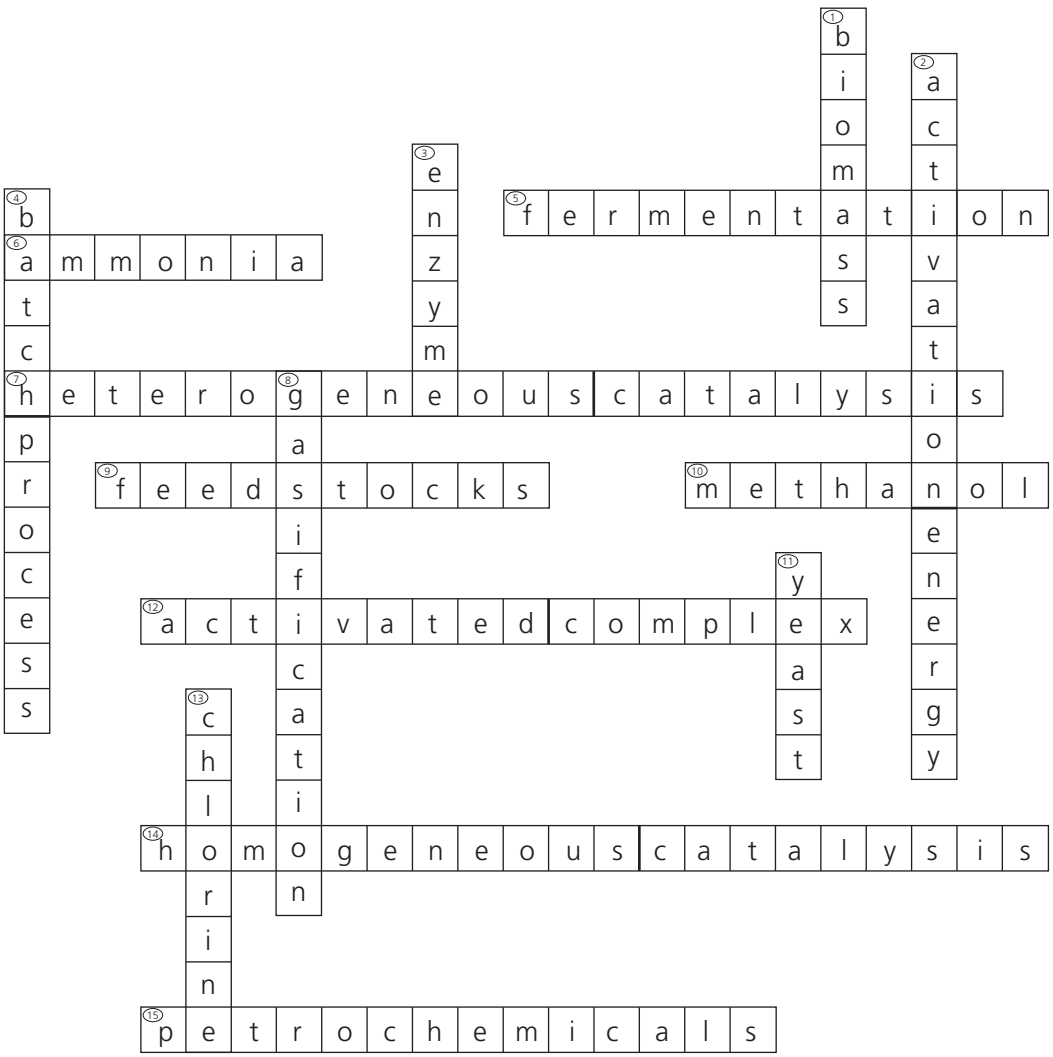
1 a) Give the THREE elements that plants need most.
Nitrogen, phosphorus and potassium
b) List the THREE main ways of fixing nitrogen from the air.
1 b) Direct combination of nitrogen and oxygen in the air during lightning flashes.
The action of bacteria.
The manufacture of ammonia by the Haber process.

2 Name the THREE main types of cells used for the electrolysis of brine in industry.
2 Main types of cells used for the electrolysis of brine in industry:
- mercury electrolytic cell;
- diaphragm cell;
- membrane cell.

3 Give the THREE main stages in the production of methanol in industry.
3 Main stages in the production of methanol in industry:
- syngas (synthesis gas) generation;
- methanol synthesis;
- crude methanol purification.

Extension

4 Write clues for the following crossword. Refer to page T49.



5 d) % by mass of nitrogen in NH_4NO_3 $= \frac{2 \times 14.0}{2 \times 14.0 + 4 \times 1.0 + 3 \times 16.0} \times 100\%$
$= 35.0\%$

% by mass of nitrogen in $CO(NH_2)_2$ $= \frac{2 \times 14.0}{12.0 + 16.0 + 2 \times 14.0 + 4 \times 1.0} \times 100\%$
$= 46.7\%$

$\therefore$ urea ($CO(NH_2)_2$) is the more effective fertilizer.

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

5 There are four main steps in the manufacture of the fertilizer ammonium nitrate.

Step 1 The reaction of methane with steam to produce hydrogen.
Step 2 The reaction of hydrogen and nitrogen to produce ammonia.
Step 3 The production of nitric acid from ammonia.
Step 4 The reaction of ammonia with nitric acid to produce ammonium nitrate.

a) Write a chemical equation for the reaction of methane with steam.
$CH_4(g) + H_2O(g) \longrightarrow 3H_2(g) + CO(g)$

b) The first stage in the product of nitric acid from ammonia is the oxidation of ammonia to nitrogen monoxide. Write a chemical equation for the reaction involved.
$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$

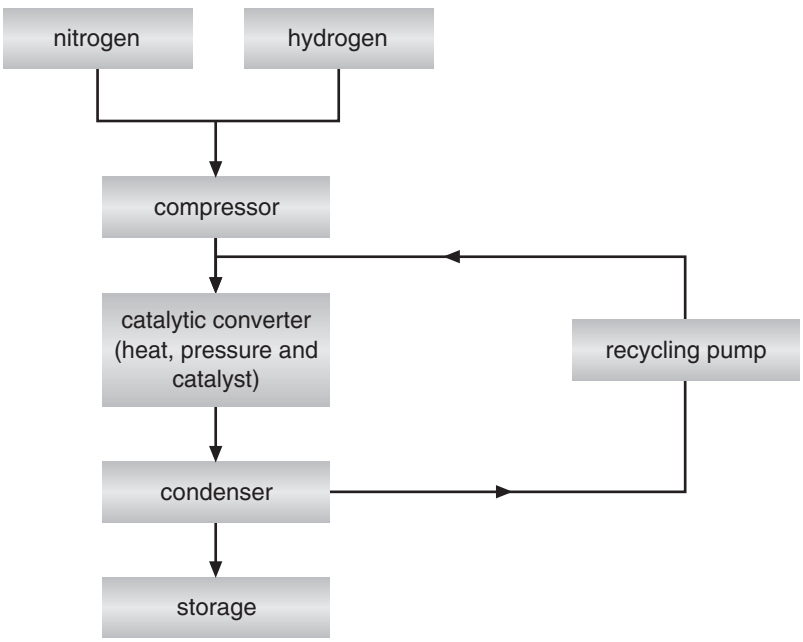
c) What type of reaction is taking place in *Step 4*?
Neutralization

d) Assume that the effectiveness of a fertilizer is directly proportional to its percentage by mass of nitrogen. Determine by calculation whether ammonium nitrate (NH_4NO_3) or urea ($CO(NH_2)_2$) is the more effective fertilizer.

(Relative atomic masses: H = 1.0, C = 12.0, N = 14.0, O = 16.0)

Exercise

6 During World War I, a German chemist, Haber, developed the following process as a source of nitrogen compounds.



a) Describe how nitrogen is obtained from the atmosphere.
By fractional distillation of liquid air

b) Write a chemical equation for the reaction between nitrogen and hydrogen in the Haber process.
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

c) Name a catalyst for the reaction.
Iron

d) What is the function of

i) the condenser?

To condense the ammonia gas into liquid.

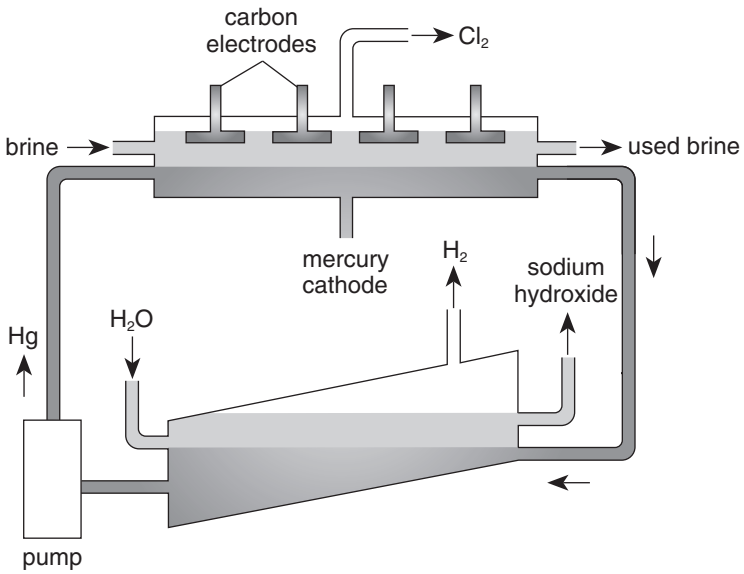
ii) the recycling pump?

Pump the unreacted nitrogen and hydrogen back to the catalytic converter to react again.

e) During the World War I, nitrogen compounds were in great demand. Suggest TWO possible reasons.

For the manufacturing of fertilizers and explosives.

7 Sodium hydroxide can be manufactured by the electrolysis of brine in the following set-up.



a) Explain which electrode, carbon or mercury, is the cathode.

b) Explain why sodium hydroxide can be manufactured by the above electrolytic process. Write equations for the reactions involved.

c) State ONE potential hazard associated with this electrolytic process.

7 a) The mercury electrode is the cathode because sodium is produced at the mercury electrode.

b) Sodium is produced at the mercury cathode. The sodium dissolves in mercury to form an amalgam. The amalgam then flows into a second cell and reacts with water to form hydrogen gas and sodium hydroxide solution. Mercury is recovered.

$2Na/Hg(l) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g) + 2Hg(l)$

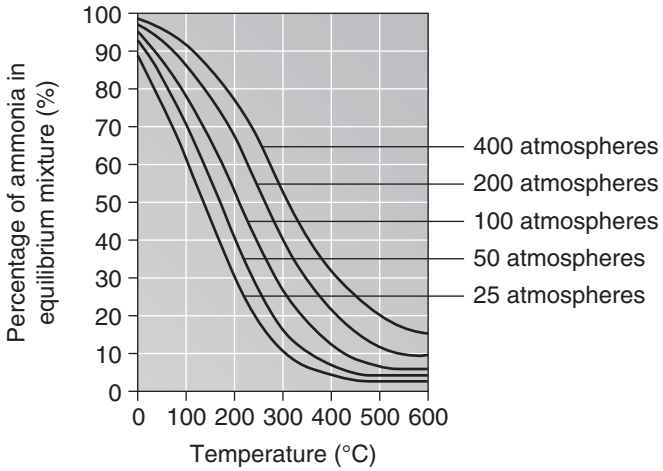
sodium amalgam

c) The process produces waste which contains poisonous mercury compounds. These waste products will cause serious pollution problems if they are discharged into rivers and seas. The marine animals may die of mercury poisoning.

8 Ammonia is manufactured by the Haber process. In the process, a mixture of nitrogen and hydrogen is passed over a heated catalyst in the reaction chamber. Nitrogen and hydrogen react according to the equation: Refer to page T50.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The graph below shows the variation of yield of ammonia with temperature at different pressures.



a) As air contains about 78% of nitrogen by volume, can air be used instead of nitrogen in the reaction? Explain.

b) Suggest how hydrogen is obtained from natural gas. Write a chemical equation for the reaction involved.

c) Deduce from the graph whether the production of ammonia from nitrogen and hydrogen is an exothermic or an endothermic reaction. Explain your answer.

d) i) State the temperature range used in the process.
400 – 450 °C
ii) Justify the temperature range used.

e) i) State the pressure used in the process.
200 atmospheres
ii) Justify the pressure used.

f) i) Identify the catalyst used in the Haber process.

Extension

ii) Is the catalyst a homogeneous catalyst or a heterogeneous catalyst? Explain.

iii) Sketch labelled energy profiles for the uncatalyzed and catalyzed reactions.

g) In the process, only about 15 % of the nitrogen is converted to ammonia. Suggest how the gaseous mixture that emerges from the reaction chamber can be treated so as to reduce the wastage of feedstocks.

9 The table below lists some information about three nitrogen-containing compounds which may be used as fertilizers. Refer to page T51.

Compound	Percentage by mass of nitrogen in compound	Solubility in water at 25 °C (mol dm^{-3})
NH_4NO_3	35.0%	26.8
$(NH_4)_2SO_4$	*x*	5.8
$(NH_4)_2HPO_4$	21.2%	4.4

a) Explain why nitrogen is essential for the growth of plants.

b) Calculate the value of *x*.

21.2%

c) The growth of some plants such as turnips requires a large amount of nitrogen. Which of the three compounds is the most suitable for use as a fertilizer for these plants? Explain your choice.

d) Ammonia reacts with sulphuric acid to give the fertilizer ammonium sulphate. Write an equation to represent the reaction.

e) The fertilizer ammonium nitrate is very soluble in water. Describe TWO advantages and TWO disadvantages of this.

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, S = 32.1)

Extension

10 Consider the manufacture of fertilizers in large production plants.

a) Give THREE factors that must be considered when making decisions about the amounts of raw materials to store.

b) Explain why most continuous manufacturing processes involve materials in the liquid or gaseous state. In continuous manufacturing processes, there must be a steady flow of reagents into the reactor and finished products out of the reactor.
It is easier for materials in liquid or gaseous state to move in and out the reactor.

c) Fertilizers are hygroscopic and tend to 'cake'. Suggest how manufacturers overcome this caking problem. By producing fertilizers which are protected, either by external coatings or as a result of their internal compositions.

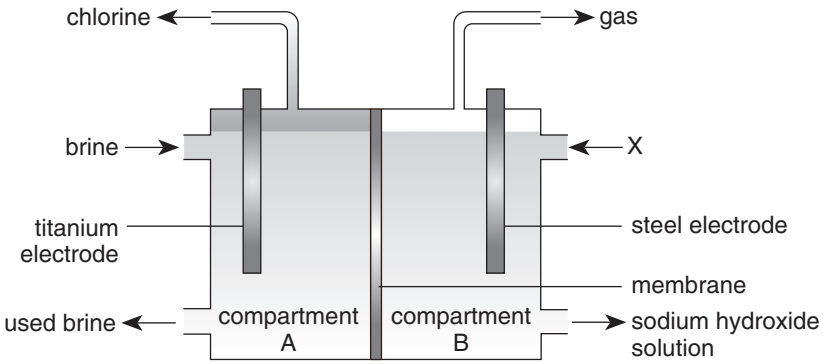
d) A newspaper article in the local newspaper of Town X suggested that fish stocks in the sea were increasing because farmers could no longer afford to buy fertilizers.

Explain how the use of nitrate fertilizers on land may have reduced fish stocks.

10 a) Any three of the following:
- The cost of the storage tanks.
- The value of the land required for storage.
- The value of the materials stored and the working capital tied up with them.
- Some materials may require special storage conditions.
- Additional safety concerns where materials are stored in bulk.
- Economies of scale when buying in bulk.
- Buy raw materials just when required rather than keeping stock.
- The cost to the company should stocks run out and production be brought to a halt.

d) Fertilizers get into the sea / fertilizers are soluble.
The fertilizers encourage algae growth. Algae grow in great green-cloudy masses so that sunlight is cut off from the other water plants.
The plants die and algae are short-lived too. Bacteria cause these plants to decompose and so use up oxygen in the water.
Fishes die due to lack of oxygen.

11 A chemical plant manufactures chlorine and sodium hydroxide by the electrolysis of brine in the following set-up.



a) Explain which electrode, titanium or steel, is the anode.
Titanium electrode

b) What type of ions can pass through the membrane?
$Na^+(aq)$ ions

c) Name substance X that goes into compartment B.
Water

d) Using the concept of preferential discharge of ions, explain the electrode reactions and why sodium hydroxide can be manufactured by the above electrolysis.

e) Suggest why this type of cells has replaced mercury electrolytic cells in many countries.

f) The chlorine produced is stored in metal cylinders. These cylinders will be used in water treatment plants.

i) Which of the following hazard warning labels should be displayed on the metal cylinders?
B



ii) Explain why chlorine is used in water treatment plants.
As chlorine is toxic, it can kill the bacteria.

g) Suggest ONE product, apart from chlorine and sodium hydroxide, likely to be manufactured in the chemical plant. Hydrochloric acid

11 d) At the titanium electrode:
A hydroxide ion is a stronger reducing agent than a chloride ion. Thus, hydroxide ions should be discharged preferentially. However, at higher concentrations of chloride ions, the discharge of chloride ions becomes more favourable.
Chloride ions are preferentially discharged (oxidized) to form chlorine gas.
At the steel electrode:
A hydrogen ion is a stronger oxidizing agent than a sodium ion. Therefore hydrogen ions are preferentially discharged (reduced) to form hydrogen gas.
Water dissociates continuously to replace the hydrogen ions discharged at the cathode. Thus, there is a build-up of hydroxide ions near to the cathode.
The membrane allows sodium ions to pass through. Thus, sodium hydroxide is produced at the cathode.
e) Mercury electrolytic cells produce waste which contains poisonous mercury compounds. These waste products cause serious pollution problems if they are discharged into rivers and seas. The marine animals may die of mercury poisoning.

12 Methanol can be manufactured from carbon monoxide and hydrogen according to the following equation: Refer to page T51.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \qquad \Delta H = -129 \text{ kJ mol}^{-1}$

The yield of methanol is 10% when the following set of operating conditions is used:

temperature – 300 °C;
pressure – 100 atm;
catalyst – copper and zinc oxide.

a) The first stage in methanol production is the conversion of methane into syngas.

i) Write an equation for the reaction involved.

ii) Give conditions required for the reaction.

b) With the intention of increasing the yield of methanol, a student suggested three other sets of conditions shown below:

Suggested set of conditions	Temperature (°C)	Pressure (atm)	Catalyst
A	300	1	Cu and ZnO
B	100	100	Cu and ZnO
C	300	100	Cr_2O_3

For each suggested set of conditions, decide and explain whether it can increase the yield of methanol.

c) Methanol can be produced from biomass.

i) Explain the meaning of the term 'biomass'.

Extension

ii) Briefly describe how methanol can be produced from biomass.

iii) Suggest ONE advantage of applying this process.

d) State TWO important uses of methanol.

13 This question is about the manufacture, properties and uses of methanol. Refer to page T52.

Methanol is manufactured from synthesis gas, a mixture of carbon monoxide, carbon dioxide and hydrogen.

80 years ago, plants making methanol used pressures of about 200 atmospheres and a temperature of 400 °C.

Today, modern plants operate at a pressure of 100 atmospheres, a temperature of 270 °C and with a copper catalyst.

a) Suggest TWO possible sources of synthesis gas from raw materials containing both carbon and hydrogen.

b) In the modern plant, what is the operating temperature in Kelvin?
543 K

c) The main reactions involved in the conversion of synthesis gas to methanol in both methods are

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ $\Delta H = -91$ kJ mol^{-1}

$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g)$ $\Delta H = -49$ kJ mol^{-1}

Compare the old and new methods by considering their operating conditions. Discuss ONE advantage of the old method and THREE advantages of the new method. Justify your answers.

d) One use of methanol is as a fuel.

i) Write the equation, including state symbols, for the complete combustion of methanol.

ii) Suggest ONE reason why methanol might be preferable to petrol as a fuel.

(Edexcel Advanced GCE (Nuffield), Jun. 2007)

Extension

14 One stage in the manufacture of sulphuric acid is the exothermic reaction Refer to page T53.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

a) In a closed container this mixture of gases would be in dynamic equilibrium. State the meaning of the words 'dynamic' and 'equilibrium' in this context.

b) i) State the conditions of temperature and pressure used industrially for the manufacture of SO_3.

ii) Justify the choice of temperature for this reaction in terms of yield and rate.

iii) The yield of products would be greater if a higher pressure were to be used for the reaction.

Suggest a reason why a higher pressure than you have given in (i) is NOT used.

c) i) Calculate ΔH for the forward reaction, given the enthalpies of formation below.
–196 kJ

	ΔH_f (kJ mol^{-1})
$SO_2(g)$	–297
$SO_3(g)$	–395
$O_2(g)$	0

ii) State why the enthalpy of formation of oxygen, $O_2(g)$, is zero.

d) i) State the formula of the catalyst used in the industrial process.

ii) Draw a diagram to show energy profiles of the uncatalyzed and catalyzed reactions.

iii) Explain how the catalyst increases the reaction rate.

Extension

e) Suggest why the sulphur trioxide produced is passed into concentrated sulphuric acid rather than water to form sulphuric acid at the end of the process.

(Edexcel Advanced Subsidiary GCE, Unit Test 2, Jan. 2006)

15 A chemical **C** is made by reacting chemical **A** with chemical **B** in a reversible reaction. **A**, **B** and **C** are all gases under the reaction conditions. Refer to page T54.

Research chemists wanted to know the optimum conditions to use in the manufacture of **C**. They carried out a series of reactions under different conditions of temperature and pressure. The percentage conversion of **A** at equilibrium is shown in the table below.

Pressure (atm)	Temperature (°C)	% A converted
10	350	8
	450	12
	550	16
20	350	11
	450	21
	550	29
40	350	18
	450	
	550	49

a) Suggest the percentage of **A** that is converted at 450 °C and 40 atm.
30 – 40%

b) i) Use the data in the above table to state the effect of increasing pressure on the percentage of **A** converted.

ii) What can be deduced, from this change, about the total number of moles of reactants **A** and **B** compared with the number of moles of product **C** in the equation for the reaction? Explain how you reached your conclusion.

c) Use the data in the above table to deduce whether the reaction between **A** and **B** is exothermic or endothermic. Explain how you reached your conclusion.

d) It was found necessary to use a catalyst in the production of **C**.

i) What is meant by a catalyst?

ii) Suggest and explain TWO reasons why catalysts are used in industrial processes.

e) Conditions were used that should have given a conversion of **A** of 39%. In the manufacture, using these conditions, it was found that only 20% conversion was achieved.

Suggest why the conversion was much less than theory suggested.

f) Give TWO economically important processes that use catalysts.

(OCR Advanced Subsidiary GCE (Salters), Chemistry of Natural Resources, Jun. 2006)

Extension

16 Based on economic and environmental considerations, discuss the pros and cons of setting up one of the following types of factory in Hong Kong:

plastics factory
or electroplating factory

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

(HKASLE 2003)

17 In 1918, Fritz Haber was awarded the Nobel Prize in Chemistry for his work on the synthesis of ammonia from its elements. Write an article on the Haber process and its importance to society.

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

(HKALE 2006)